

ADVANCED NAVIGATION & POSITIONING CORPORATION



TLS / TTLS Technician Training

489 N. 8TH ST. #203, HOOD RIVER, OREGON 97031
TELEPHONE 541.386.1747 WWW.ANPC.COM

Transponder Landing System (TLS) Technician Course

By the end of this course, TLS Technicians will be able to:

- Deploy a Transportable TLS
- Install
- Calibrate
- Maintain a TLS
- Operate a TLS
- Prepare a TLS for transport

Overview:

- Lesson Length: approx. *40.0 Hrs*
- Materials Required:
 - Student Guide
 - TLS Technical Manual
- Equipment Required:
 - Transponder Landing System
 - Tool Kit
- Safety
 - Don't stick your fingers in the socket
 - Climbing harness and hardhats
 - Use two-man lift
- Admin



Introducing ANPC

- **Advanced Navigation & Positioning Corporation (ANPC)** is a privately-owned U.S. company located in Hood River, Oregon
- The Company is a recognized expert in ATC surveillance, navigation and precision landing, and has operated continuously since its establishment in 1991
- ANPC supplies Transponder Landing Systems (TLS) and has in-house capabilities to support all phases of TLS design, production, commissioning, training and after-sales service
- ANPC is ISO9001:2015 certified
- TLS is the proprietary product of ANPC
- ANPC is the world's only TLS supplier



Introducing TLS

- The TLS is a ground-based surveillance and navigation system used for precision approach and landing by civil and military aircraft

- TLS provides CAT I guidance like a conventional Instrument Landing System (ILS), but can be installed/operated at sites where ILS will not work:

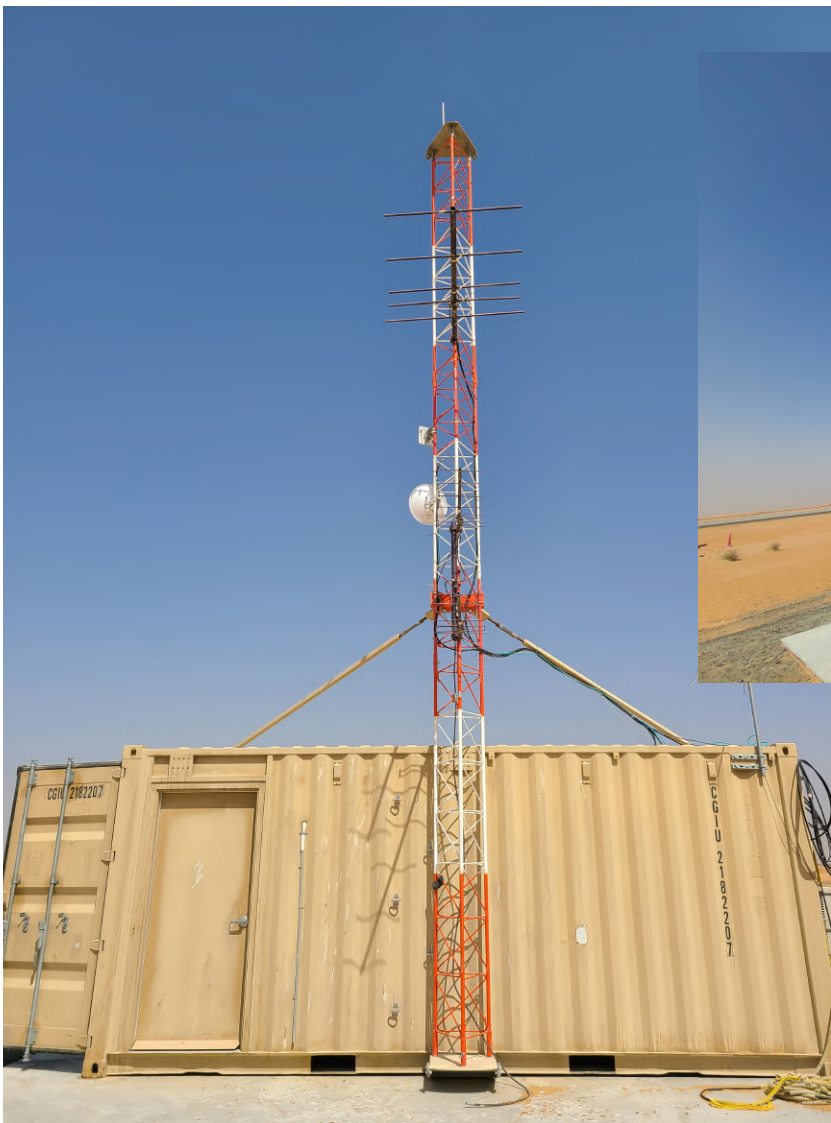
- Mountainous, coastal or other challenging terrain

- Short runways or other land availability limitations

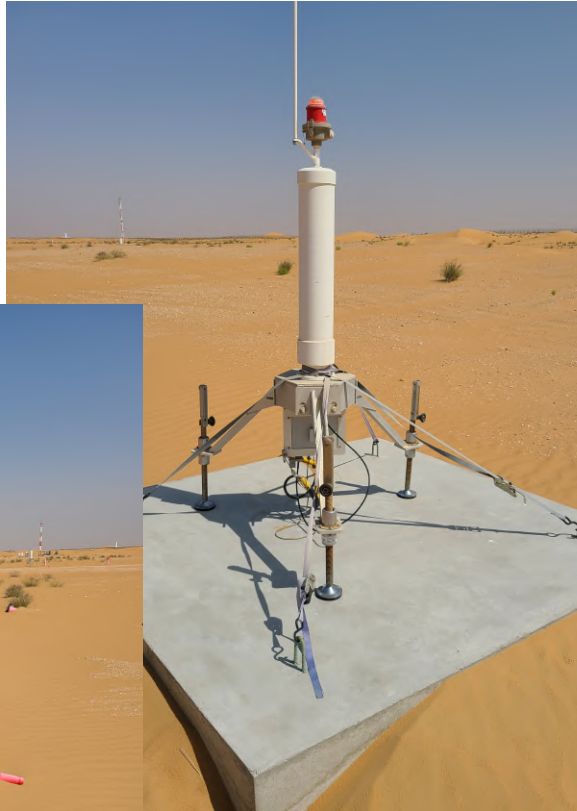
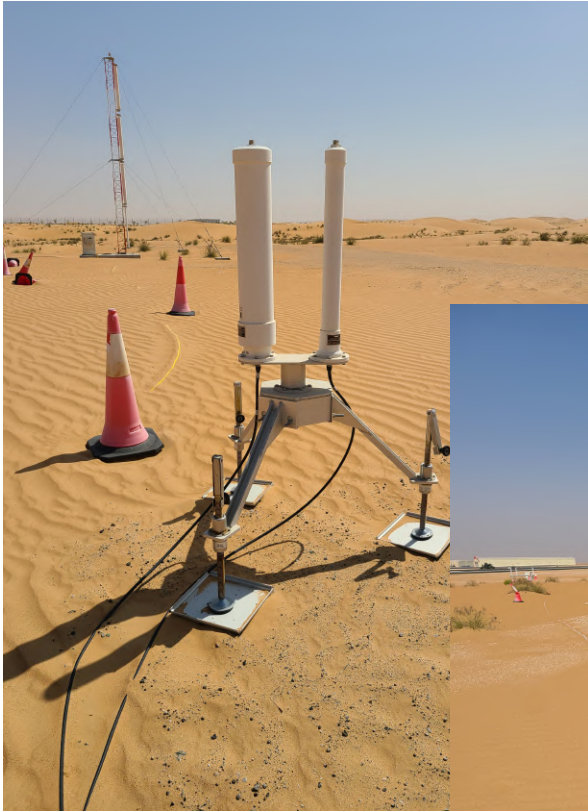
- Buildings, roads, bridges or other physical obstructions

- TLS also operates at airports where satellite-based systems (e.g., GBAS) will not work or are unreliable









TLS Technician Course Outline

- Introduction to TLS
- TLS components
- Block diagram
- Virtual emanation points
- Siting criteria
- Installation
- Configuration
 - Frequency programming
 - Integrity Monitor
 - Calibration
- Flight verification
- Maintenance
- Written test, Practical test
- Course evaluation

Documentation and study aids

- This presentation is used as a study aid to help with perspective and integration of TLS topics
- Important sources of training information include:
 - Four volumes of Technical Instruction Manual
 - 020-00071 Calibration
 - 020-00072 Operations
 - 020-00073 Installation
 - 020-00074 Maintenance
 - Assembly drawings
 - 950-00585 has list of assemblies

Documentation and study aids

- Site drawings

- 070-00xxx

- 020-00076 Deployment document

- 020-00077 Packing document

- Software:

- RCU software

- Desktop TLS software

- TLS tools

- Recorded data

Introducing TLS

- The TLS is a ground-based surveillance and navigation system used for precision approach and landing by civil and military aircraft

- TLS provides CAT I guidance like a conventional Instrument Landing System (ILS), but can be installed/operated at sites where ILS will not work:

- Mountainous, coastal or other challenging terrain

- Short runways or other land availability limitations

- Buildings, roads, bridges or other physical obstructions

- TLS also operates at airports where satellite-based systems (e.g., GBAS) will not work or are unreliable



Introducing TLS (Continued)

TLS is transportable and rapidly deployable

May be deployed to temporary or remote locations where an ILS is not technically feasible or economically practical:

Upslope or downslope under the approach path

Short runways ending at water

Temporary airfields during airport establishment or enhancement (e.g., runway extension)

Military deployments



TLS Credentials

- **Certified by the U.S. Federal Aviation Administration for CAT I operations**
- **Complies with ICAO Annex 10 standards and recommendations**
- **Meets or exceeds transportability requirements of the U.S. Air Force (USAF) Air Traffic Control, Approach and Landing Systems (ATCALS) office**
 - 4 technicians/4-hour assembly, calibration and operation**
 - 4 technicians/2.5-hour disassembly, packing and shipment-ready**
 - Fully transportable in one C-130 aircraft**
- **Operating in fixed-base and mobile applications worldwide**



TLS Operational Overview

Using proprietary software, multilateration techniques and the aircraft's beacon transponder signal used by ATC for secondary surveillance, TLS calculates the aircraft's position in 3 dimensions relative to the runway:

Range

Elevation Angle

Azimuth Angle

TLS then computes course corrections which are transmitted to the aircraft using VHF (Localizer) and UHF (Glideslope) signals identical to those of a conventional Instrument Landing System (ILS)

Course corrections are received by the aircraft and displayed to the pilot using the same equipment as a conventional ILS



TLS Operational Overview (Continued)

The visual indication directing the pilot to the runway centerline at a 3 degree rate of descent is identical to a traditional ILS landing

TLS can also generate an audible tone to emulate conventional marker beacons

TLS can support up to 4 aircraft simultaneously

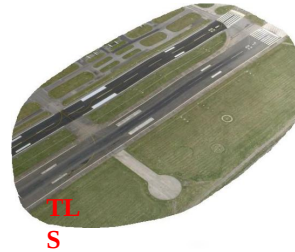


TLS Operational Sequence

STEP 1: TLS interrogates all aircraft transponders



STEP 2: All aircraft Transponders Respond



STEP 3: TLS receives transponder replies at multiple antennas and digital processing determines all aircraft positions



Localizer
Eleva
tion
guide

STEP 4: TLS detects the transponder code and updates ILS Localizer and glide slope to the ILS approach path for an aircraft



TLS Components and Siting



Azimuth Sensor



Base Station and
UHF/VHF Uplink
Antenna



Interrogation
transmitter

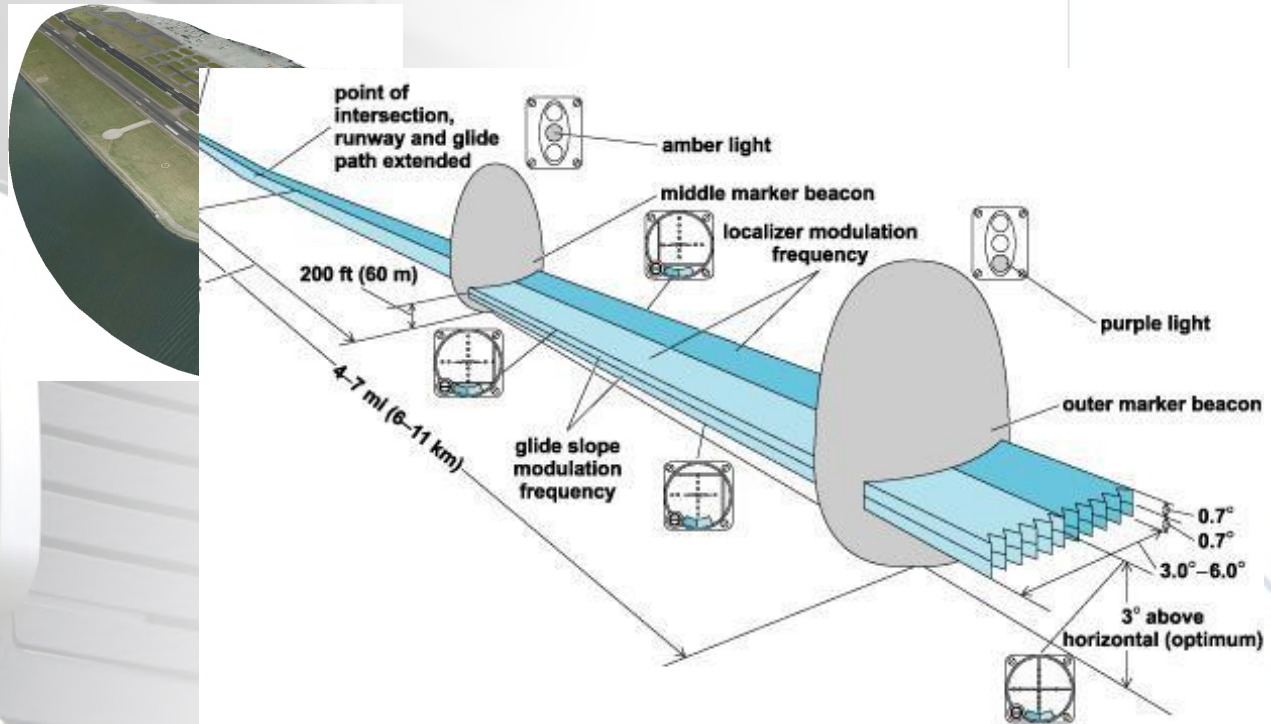


Elevation Sensor

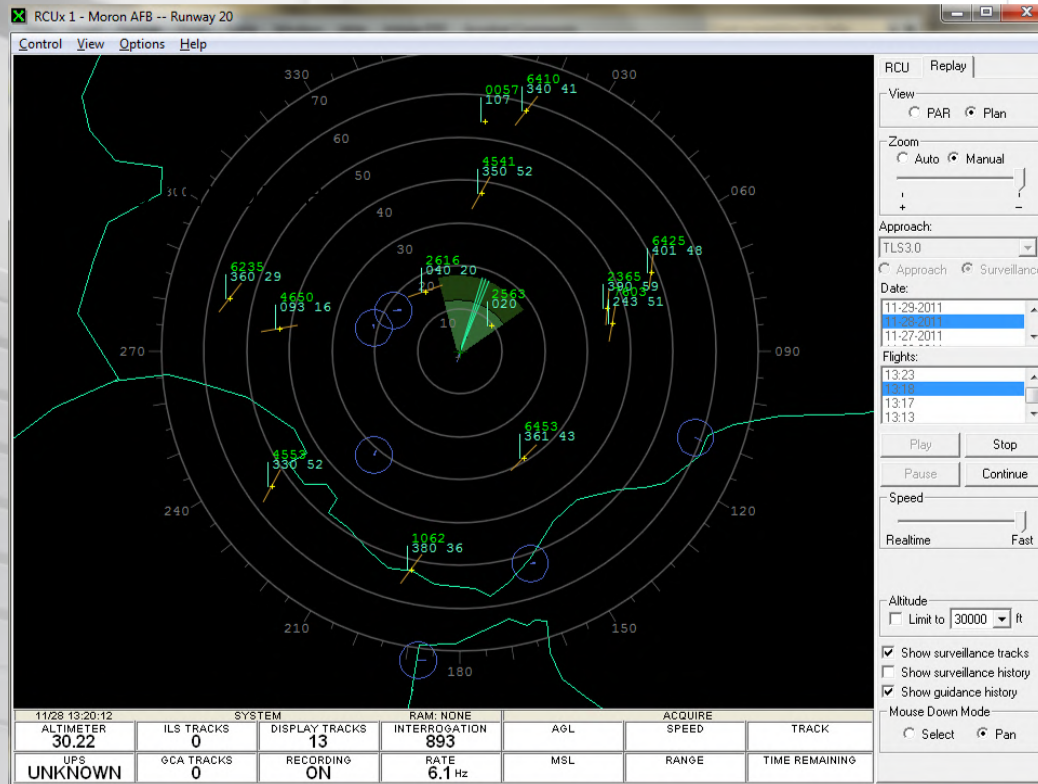


CAL/BIT

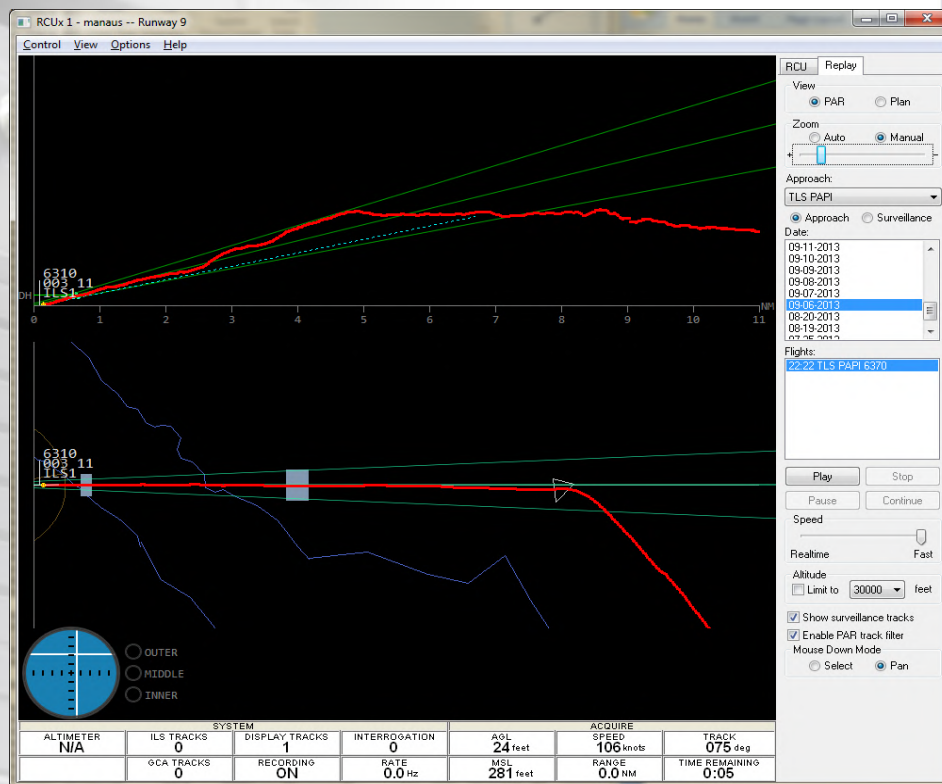
TLS Provides ILS guidance



TLS provides a secondary radar display



TLS Provides a Precision Approach Radar (PAR) display

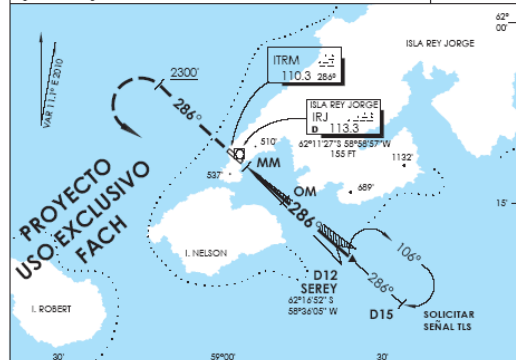


Sample TLS Instrument Approach Plate for Marsh Aerodrome, Antarctica

AD. TENIENTE RODOLFO MARSH M.
ISLA REY JORGE - ANTARTICA CHILE

SCRM IAC 6
TLS Rwy 29

APP MARSH Marsh Approach	119.7	TORRE MARSH Marsh Tower	118.1	CONTROL TERRESTRE Ground Control	
LOC ITRM 110.3	FINAL APCH CRS 286°	GS OM 1400'	DA 380'(253')	AD Elev. 147'	THR 29 Elev. 127'
APCH FRUSTRADA: Ascenso 2300' R 286°, luego viraje izquierda ascenso 3500'. Ingrese HLDG SEREY 286° derecho, o según autorización ATC. Missed Apch: Climb to 3300' on 0286, left turn climbing to 3500', enter on HLDG SEREY 286° right or according to ATC clearance.					6000 FT MSA 25 NM IRJ



2300'	RDL 286°	LT	SEREY	Nivel de Transición by ATC Transition Level by ATC	ALT MNM 3500'
DIRECTO RWY 29 Straight in RWY 29			CIRCULANDO Circling	ALTN Alternate Minimums	
DA 380' (253')					
A	1.2 Km		760' (613')	1.6 Km	
B					
C				2.8 Km	

Posibles témpanos en el área de aproximación, lecturas de Radiómetro pueden ser erráticas sobre hielo o nieve.
Possible icebergs in the approach area, readings from Radiometer can be erratic over ice or snow.

Veloc. Terrestre Kts	90	120	150	180	210
DME dist in MM y OM, sólo referencial					
DME dist at MM and OM only referential					

CAMBIO: Proyecto DGAC AMDT GRAFICADO MAR 2010

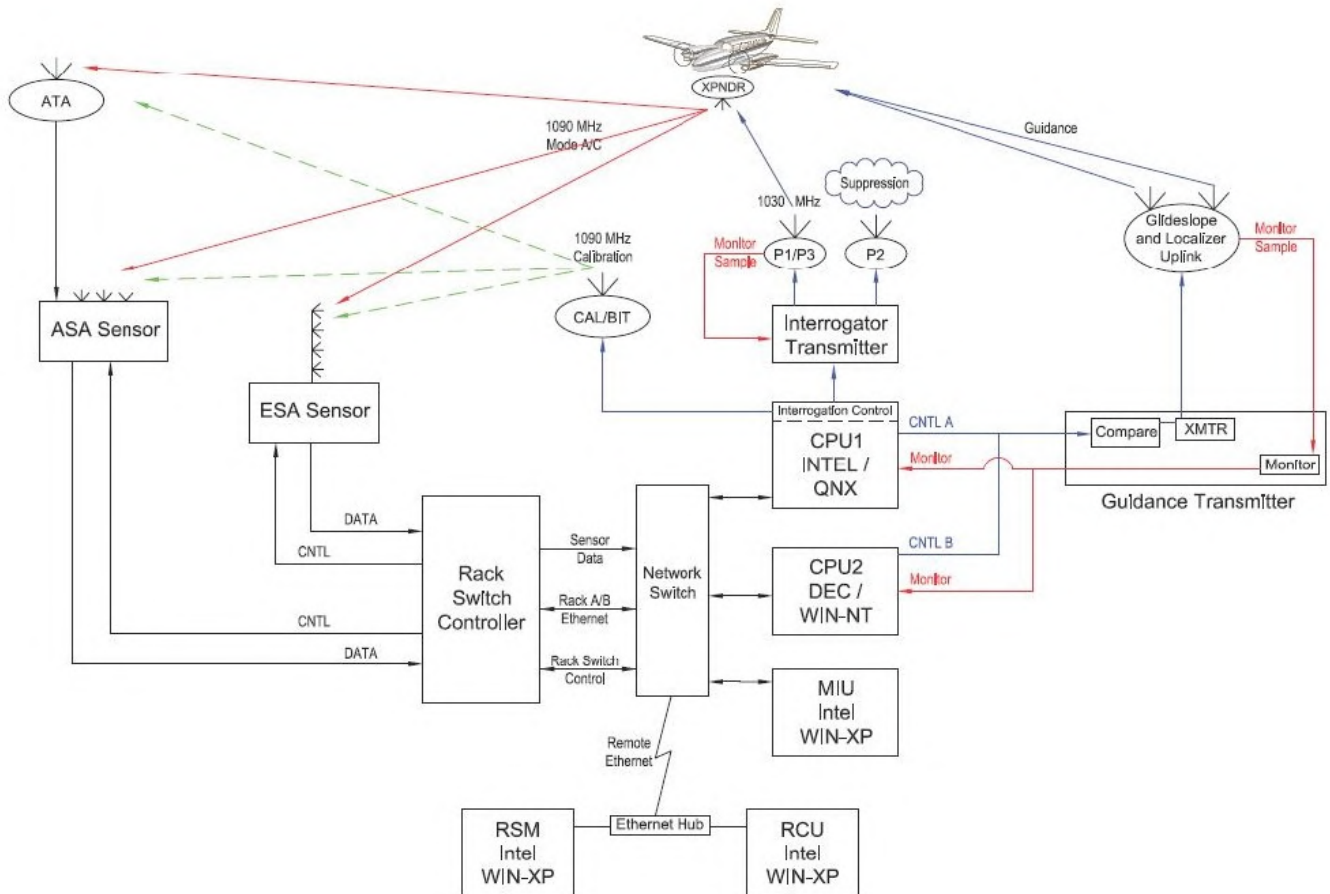
[illegible]

Technical contents

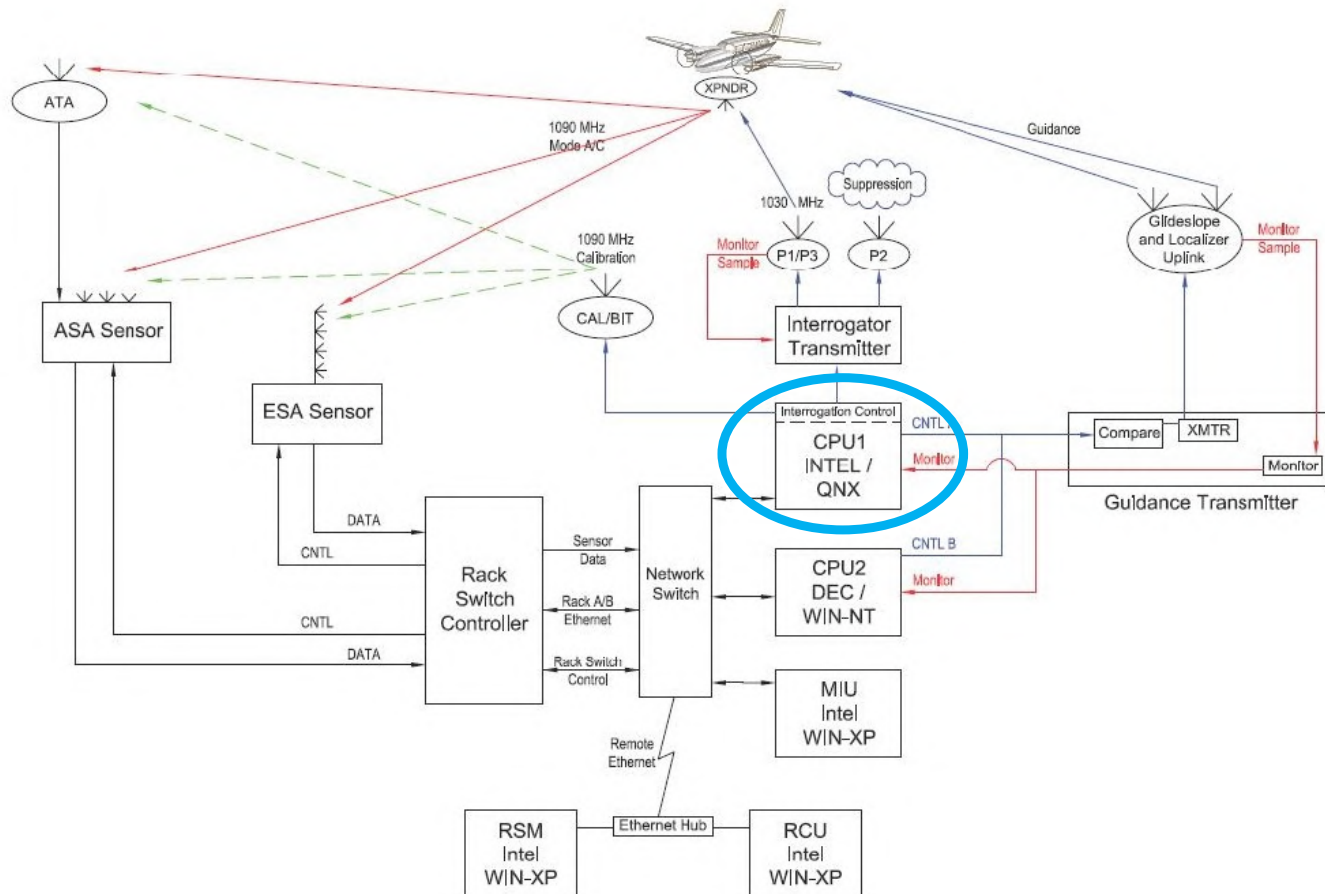
- ✓ Introduction to TLS
 - Block diagram
 - TLS components
 - Virtual emanation points
 - Siting criteria
 - Transportable TLS
 - Theory (very briefly)
 - Survey procedure
 - Configuration of the integrity monitor
 - Calibration
 - Flight verification



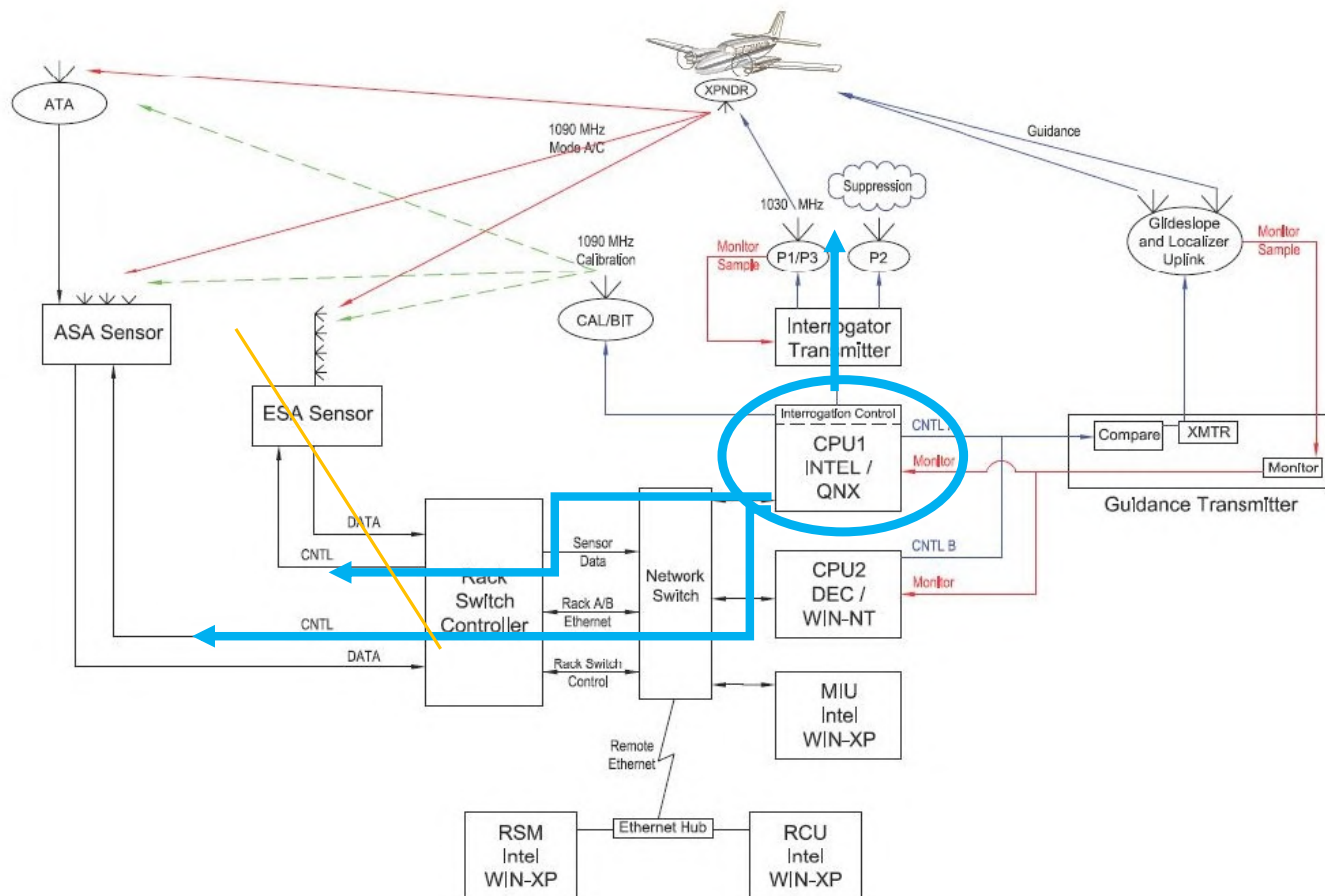
TLS System Block Diagram



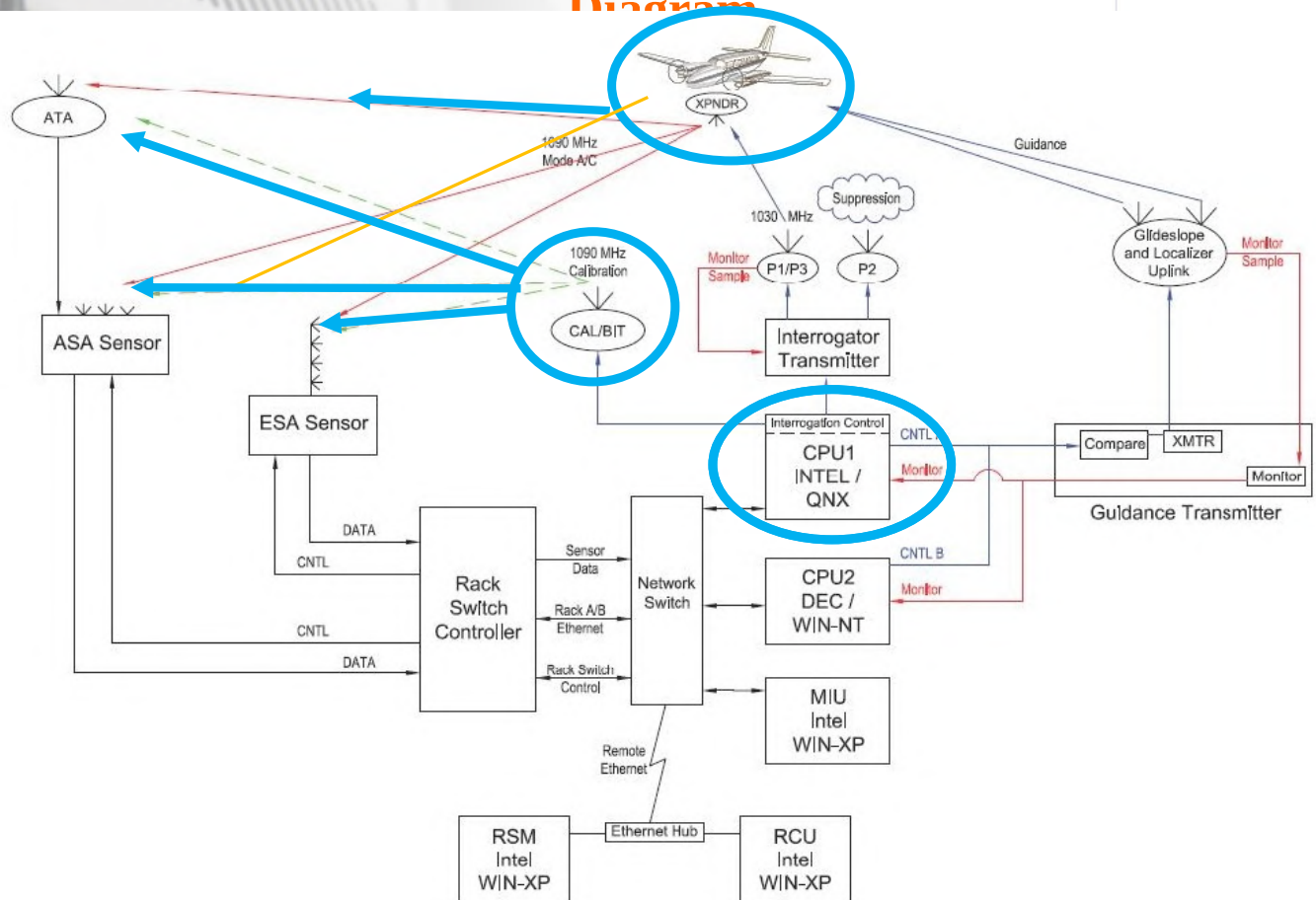
TLS System Block Diagram



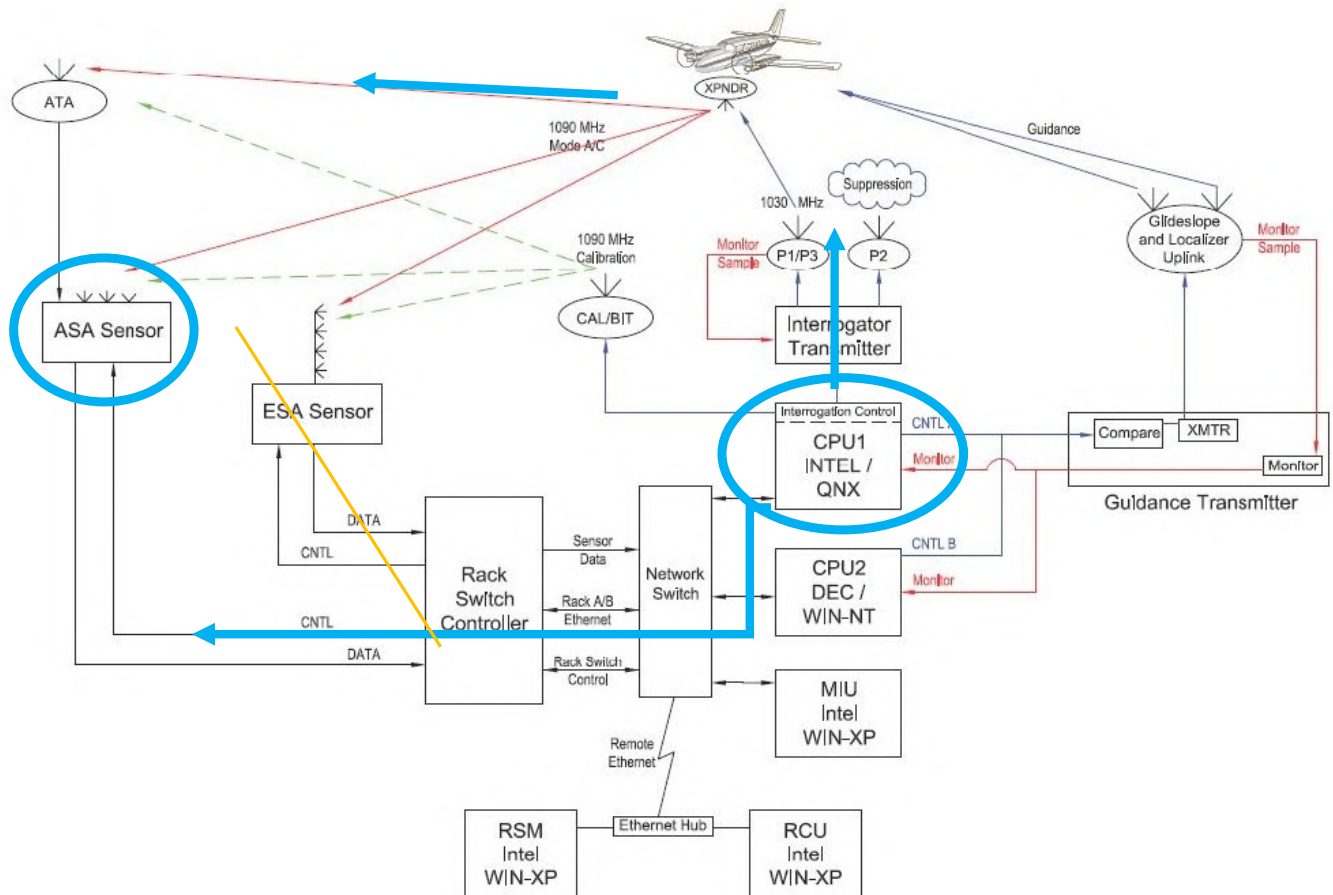
TLS System Block Diagram



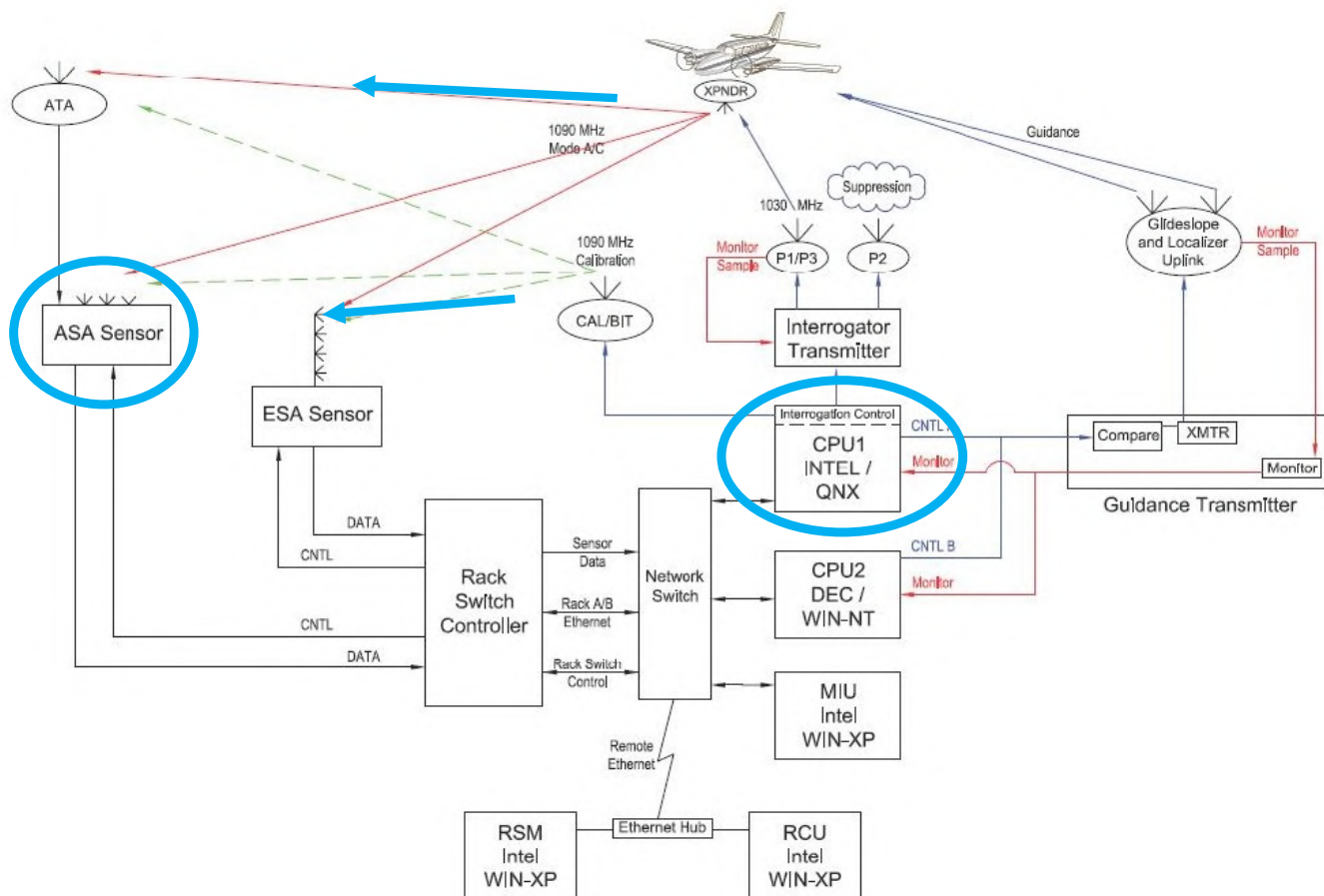
TLS System Block Diagram



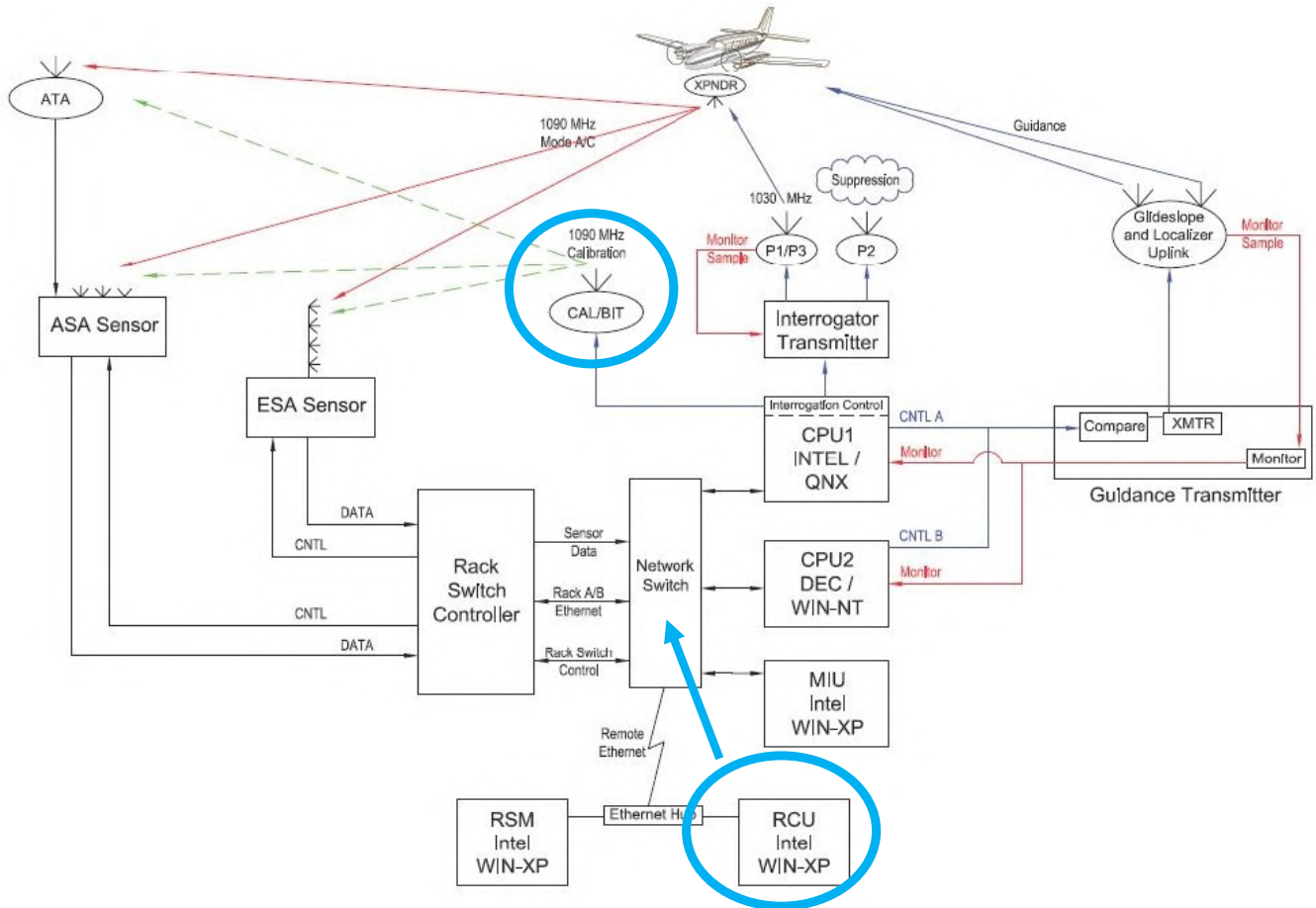
TLS System Block Diagram



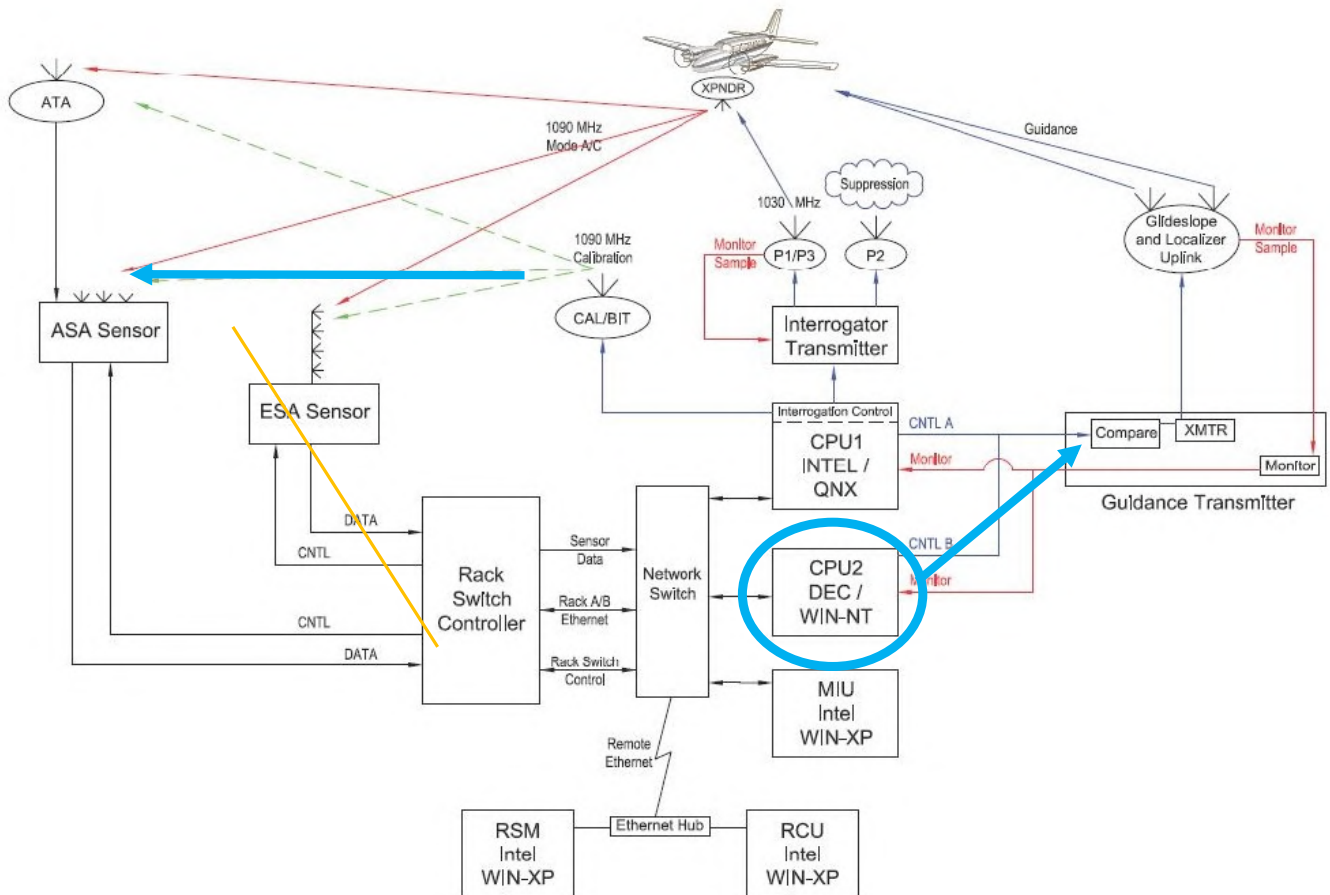
TLS System Block Diagram



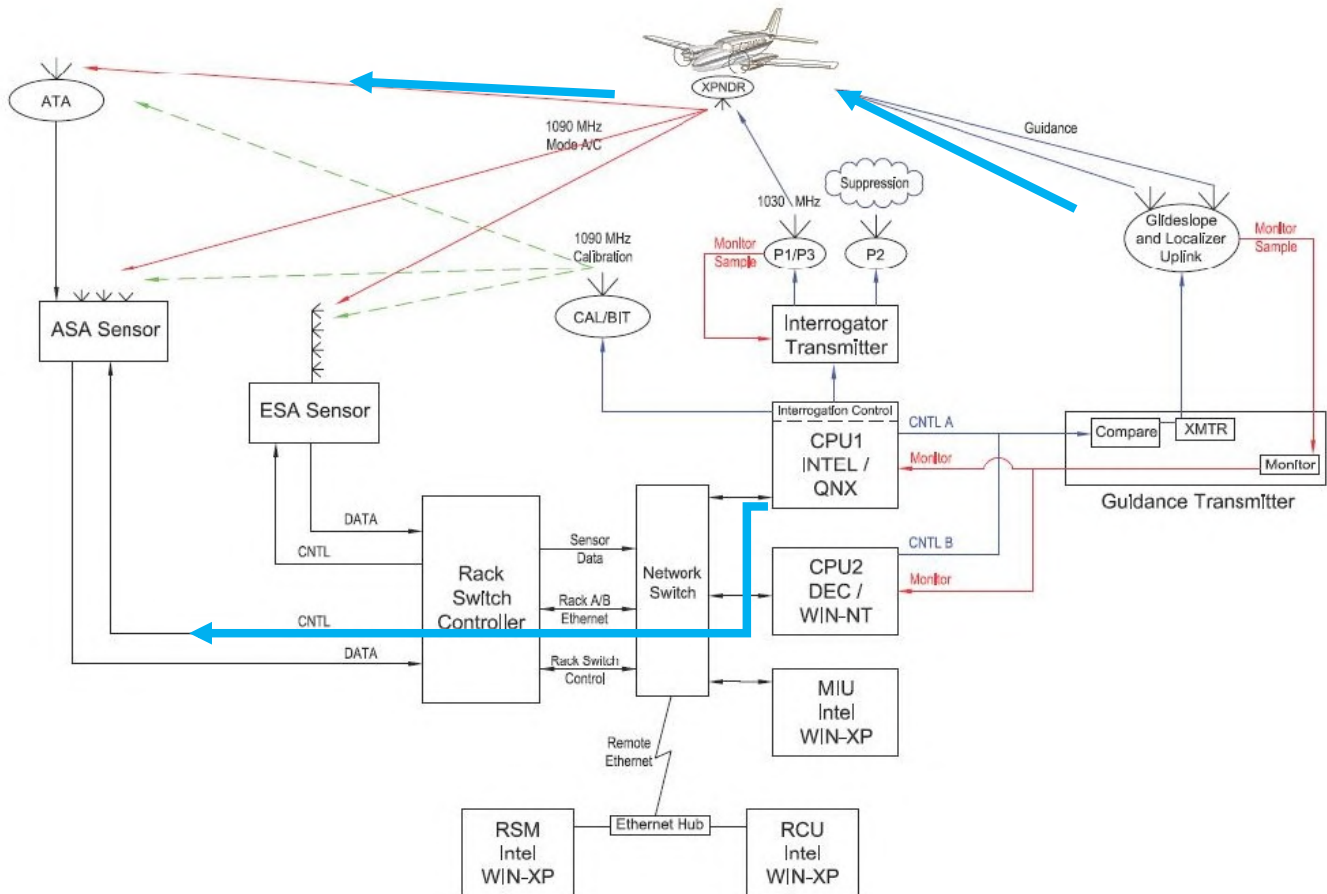
TLS System Block Diagram



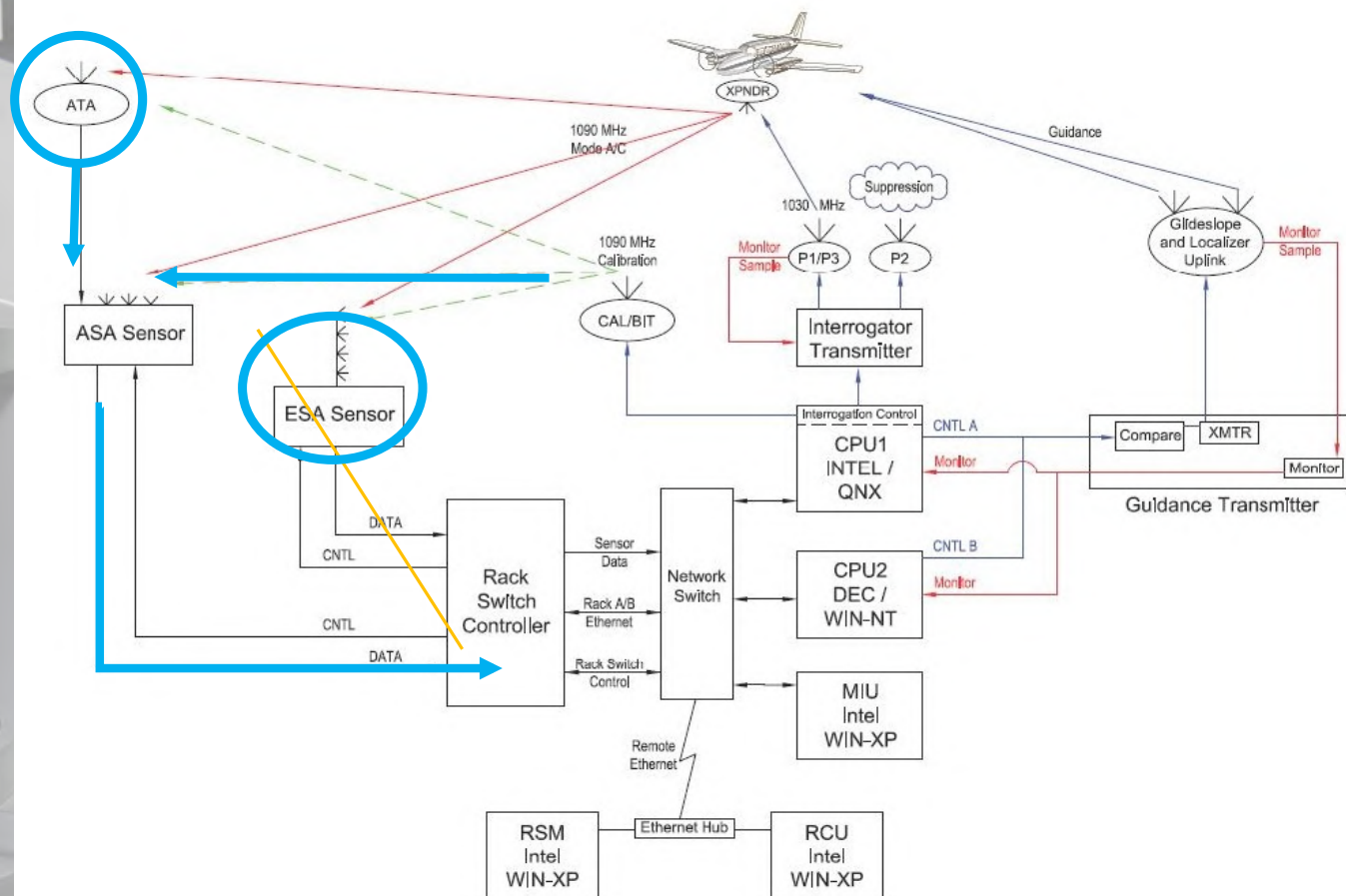
TLS System Block Diagram



TLS System Block Diagram



TLS System Block Diagram



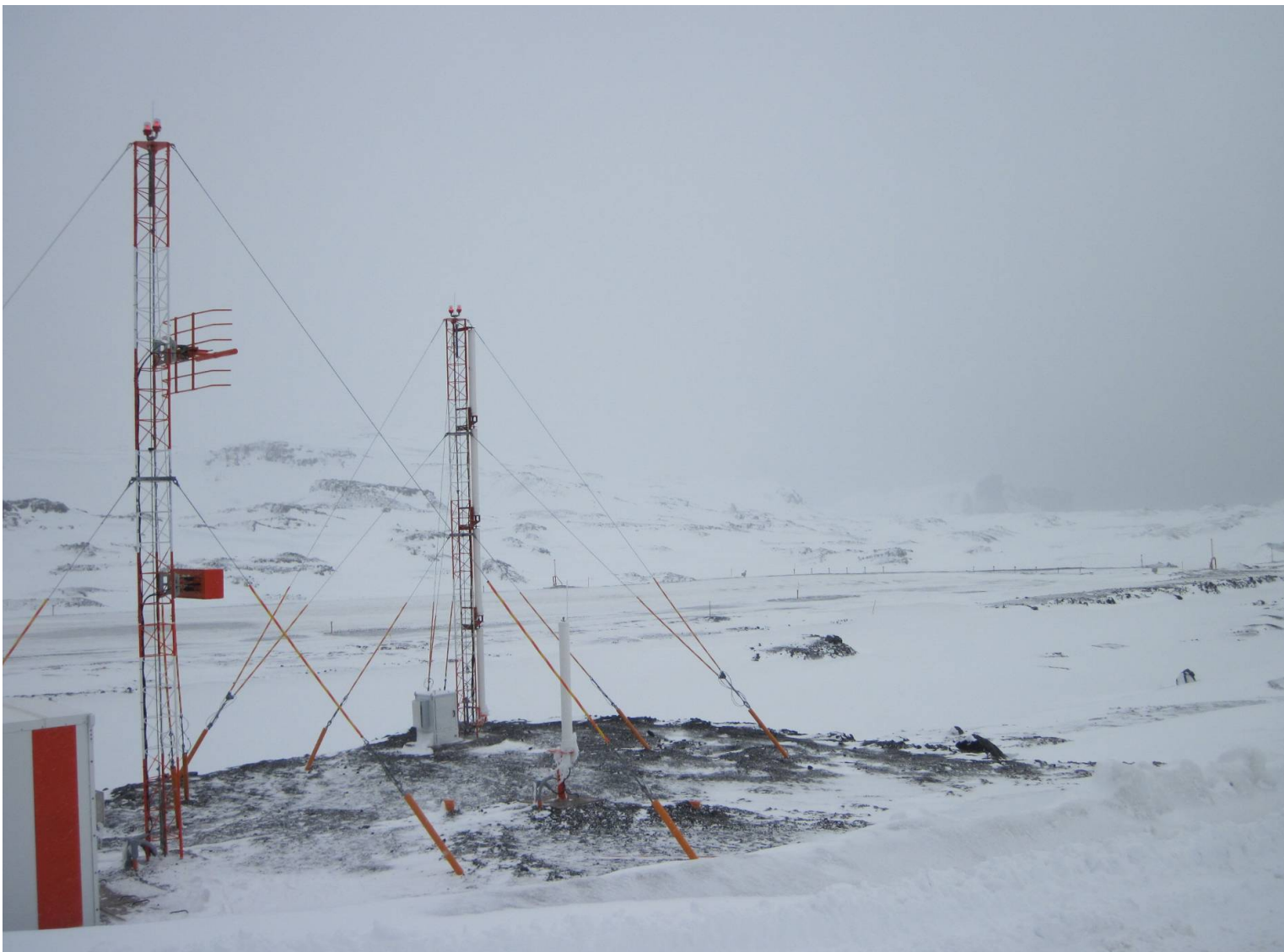
Technical contents

- ✓ Introduction to TLS
- ✓ Block diagram
- TLS components
- Virtual emanation points
- Siting criteria
- Transportable TLS
- Theory (very briefly)
- Survey procedure
- Configuration of the integrity monitor
- Calibration
- Flight verification



TLS components

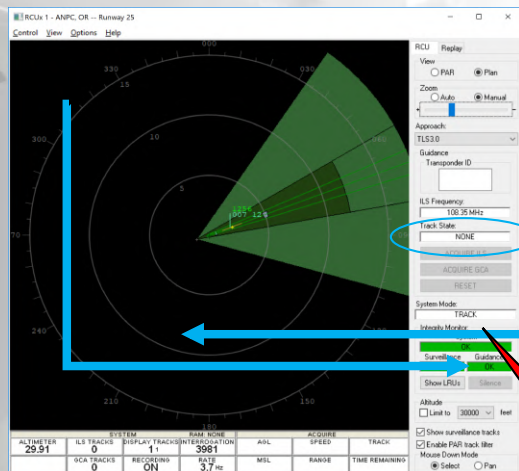




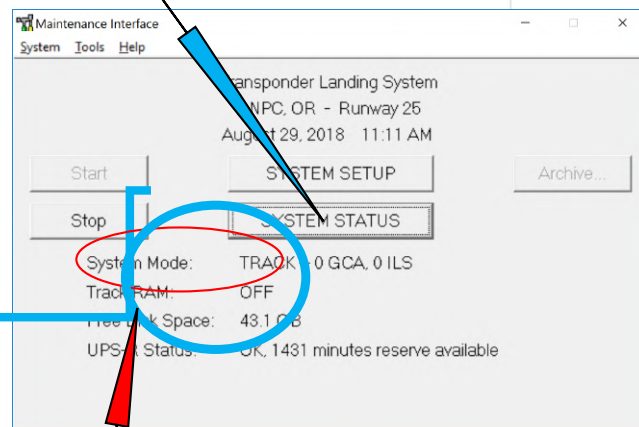


System Mode and Track State

- System Mode and Track State are displayed on the RCU and MIU

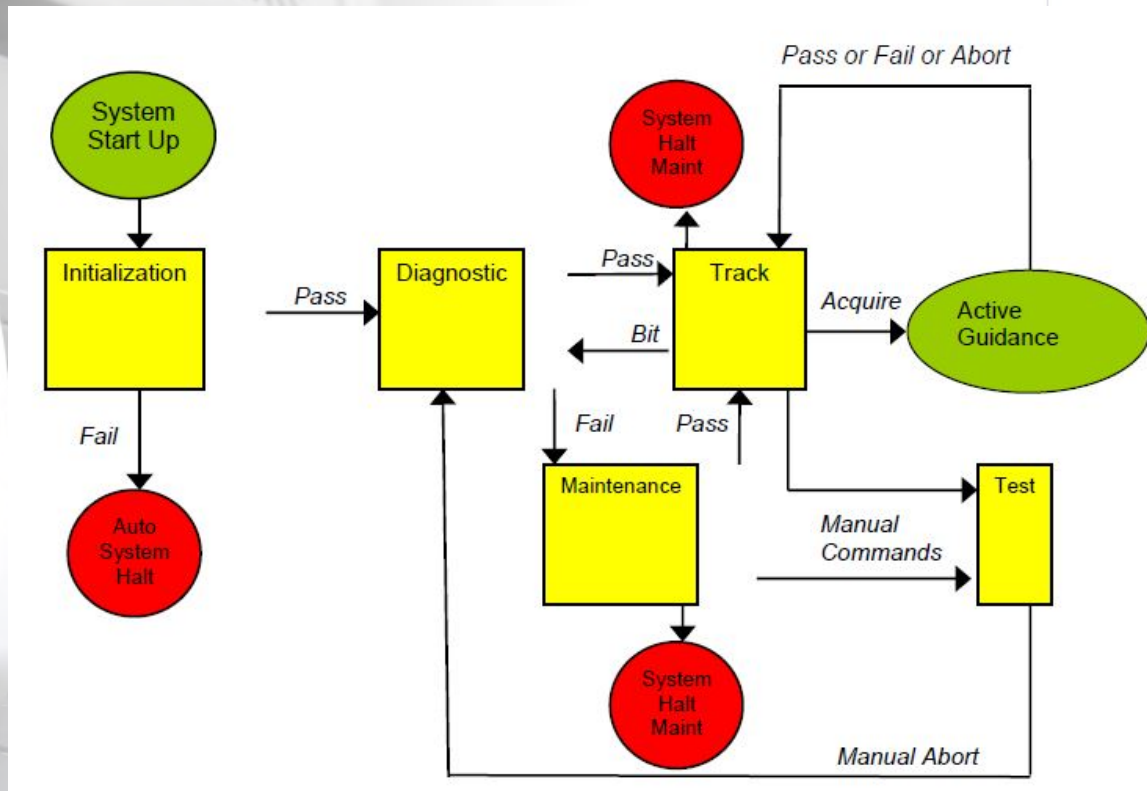


Track State



System Mode

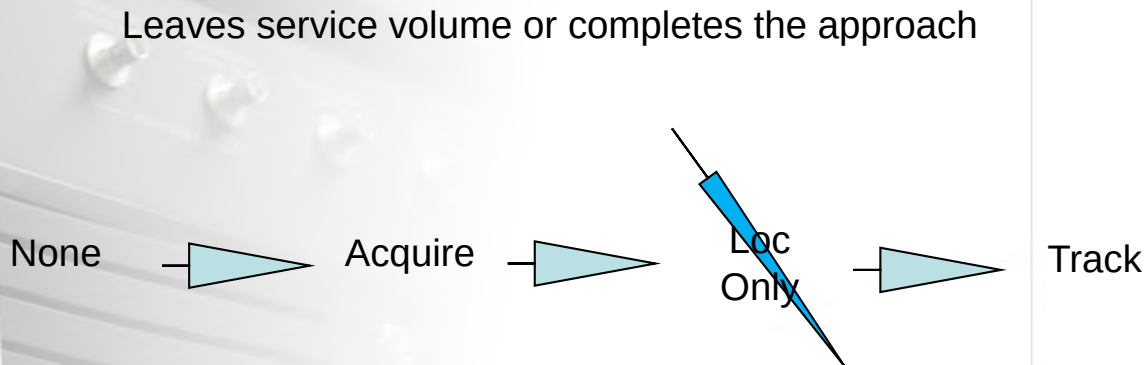
System modes



Track States

- Track States can be one of the following
 - ☐ None – the TLS is not searching for a specific transponder code
 - ☐ Acquire
 - o a transponder code has been entered on the RCU and the TLS is searching for that specific transponder code
 - ☐ Loc Only
 - o The TLS has found a match to the transponder code entered on the RCU and
 - o is transmitting localizer signal on the frequency assigned to that RCU
 - ☐ Track
 - o The aircraft is within the GS volume and
 - o the TLS is transmitting both Loc and GS

Track States

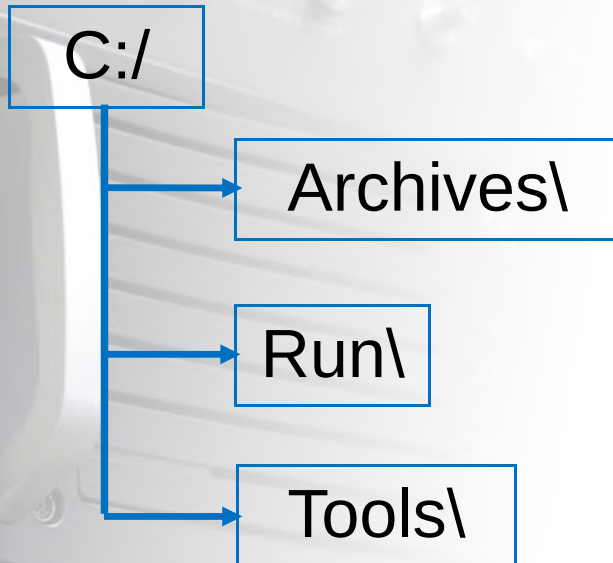


Leaves GS volume

Has a track alarm – will re-acquire

MIU Folder Structure

Windows Operating System



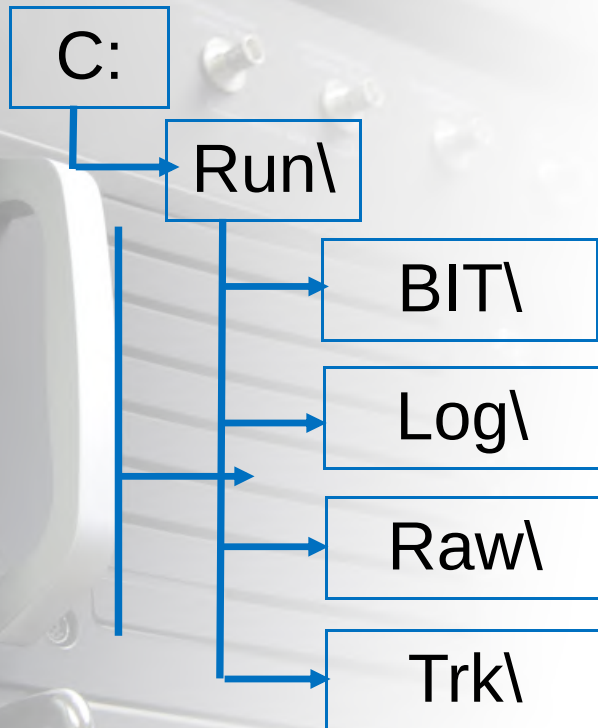
Archiving procedure stores data here

Executables and configuration files

Software tools

Base2 Folder Structure

NT Operating System



Executables and config files

BIT data

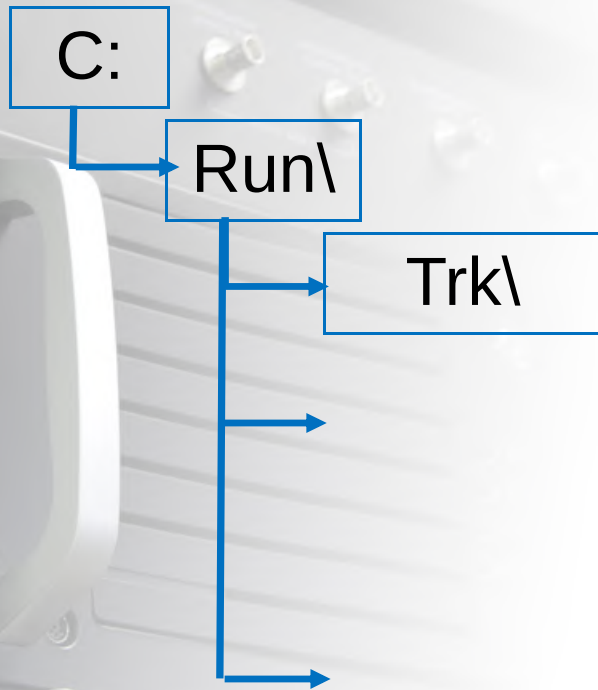
Operational log record

Approach data

BIT data during track mode

RCU Folder Structure

Windows Operating System



Executables and config files

Records of surveillance and approach data



Technical contents

- ✓ Introduction to TLS
- ✓ Block diagram
- ✓ TLS components
 - Virtual emanation points
 - Siting criteria
 - Transportable TLS
 - Theory (very briefly)
 - Survey procedure
 - Configuration of the integrity monitor
 - Calibration
 - Flight verification



TLS uses virtual emanation points

- There is no antenna located at the point from which the TLS signal appears to emanate
- This is because:

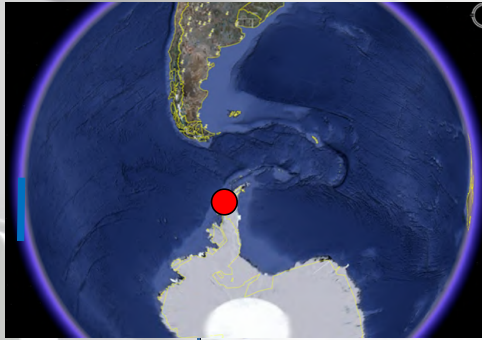
1. TLS determines the position of the aircraft based on replies from the aircraft transponder
2. TLS uses the position of the aircraft to determine the ILS signal that would exist at that location
3.  TLS transmits the signal (for that aircraft position) to the aircraft on a standard ILS frequency

Repeat 1-3 above 20 times per second

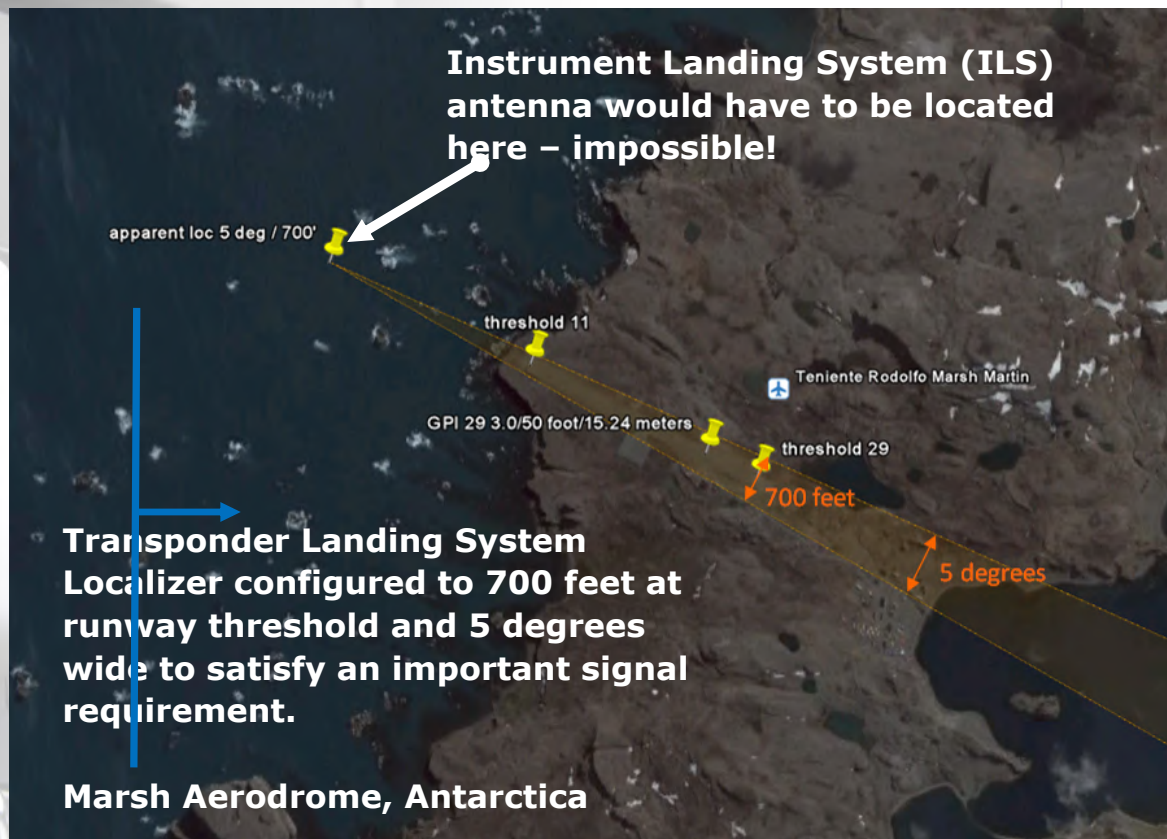
- Therefore TLS can project a “virtual emanation” point for the localizer to a position where an ILS localizer can not be placed e.g. in the water near a coastline



Benefit: Marsh Base, Antarctica



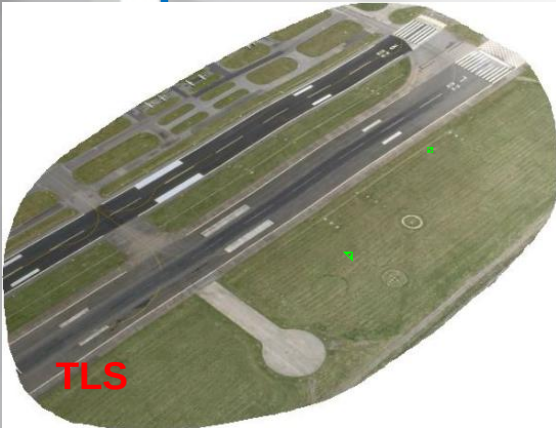
Short Runway is a Problem for Instrument Landing System



Technical Operational Sequence – 4 Steps

STEP 1:

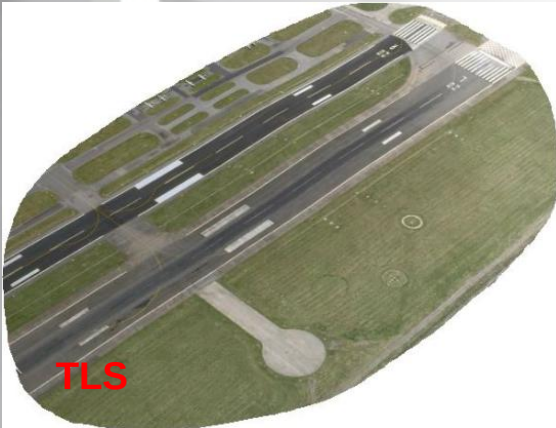
TLS interrogates all
aircraft transponders



Technical Operational Sequence

STEP 2:

All aircraft Transponders
Respond



TLS



Technical Operational Sequence

STEP 3:

TLS receives transponder replies at multiple antennas and digital processing determines all aircraft positions in x, y and z



E
l
e
v
a
t
i
o
n
Y

X



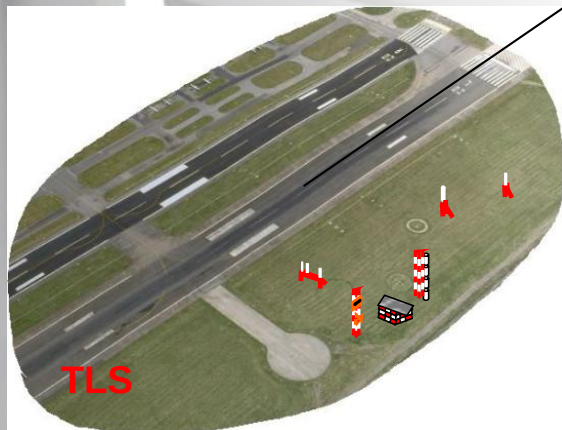
Technical Operational Sequence

STEP 4 :

TLS determines the localizer and glide slope needle deflections that correspond to the aircrafts position and transmits that signal to the aircraft

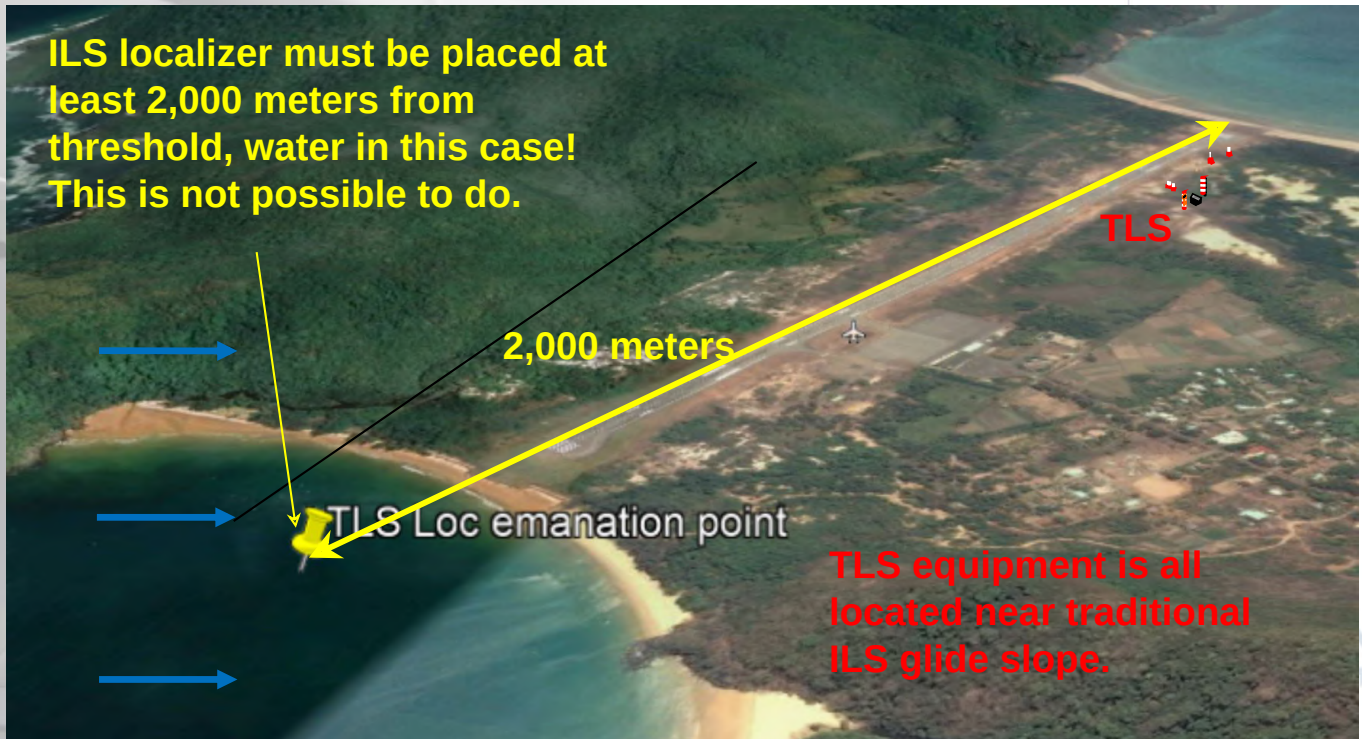


Course deviation indicator - CDI



TLS

ILS localizer equipment placement compared to TLS



ICAO Annex 10 requirement for localizer

v3 3.1.3.7.1 "The maximum course angle shall not exceed 6 degrees" and...

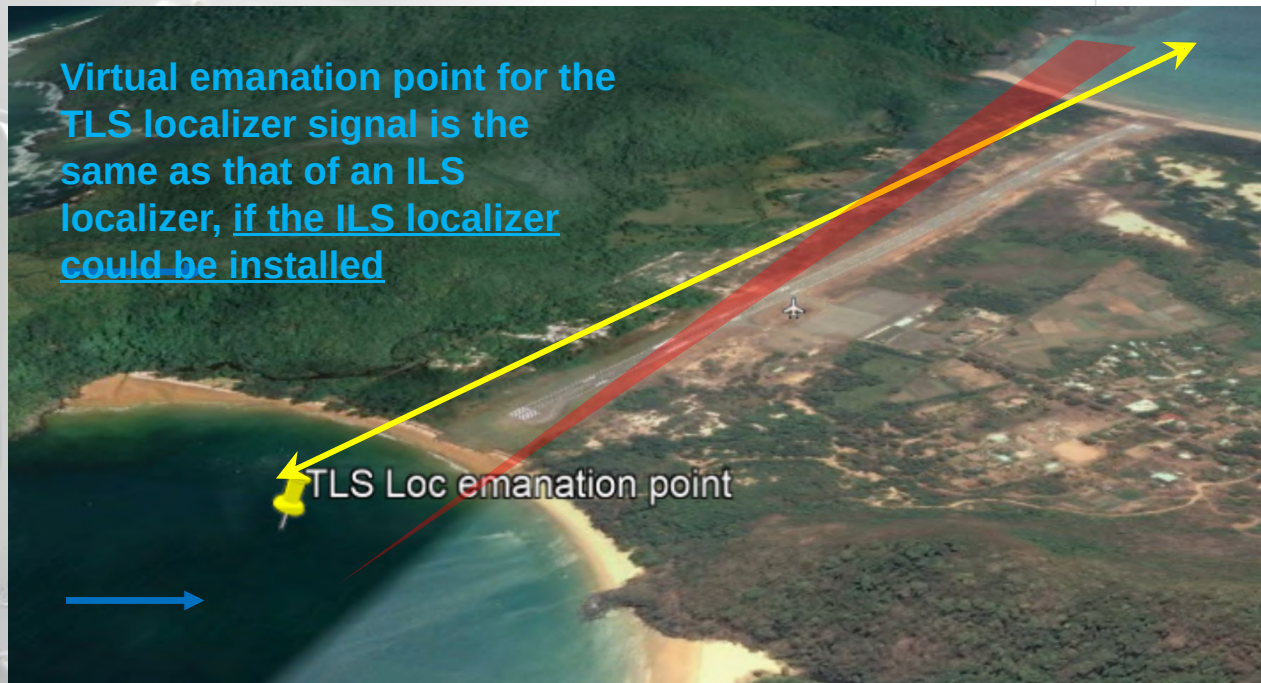
Note1: The "nominal sector width of 210 m ... at the ILS reference datum"

As a consequence of the above two ICAO requirements, an ILS localizer must be placed at least 2,000 meters from threshold



ILS localizer signal compared to TLS

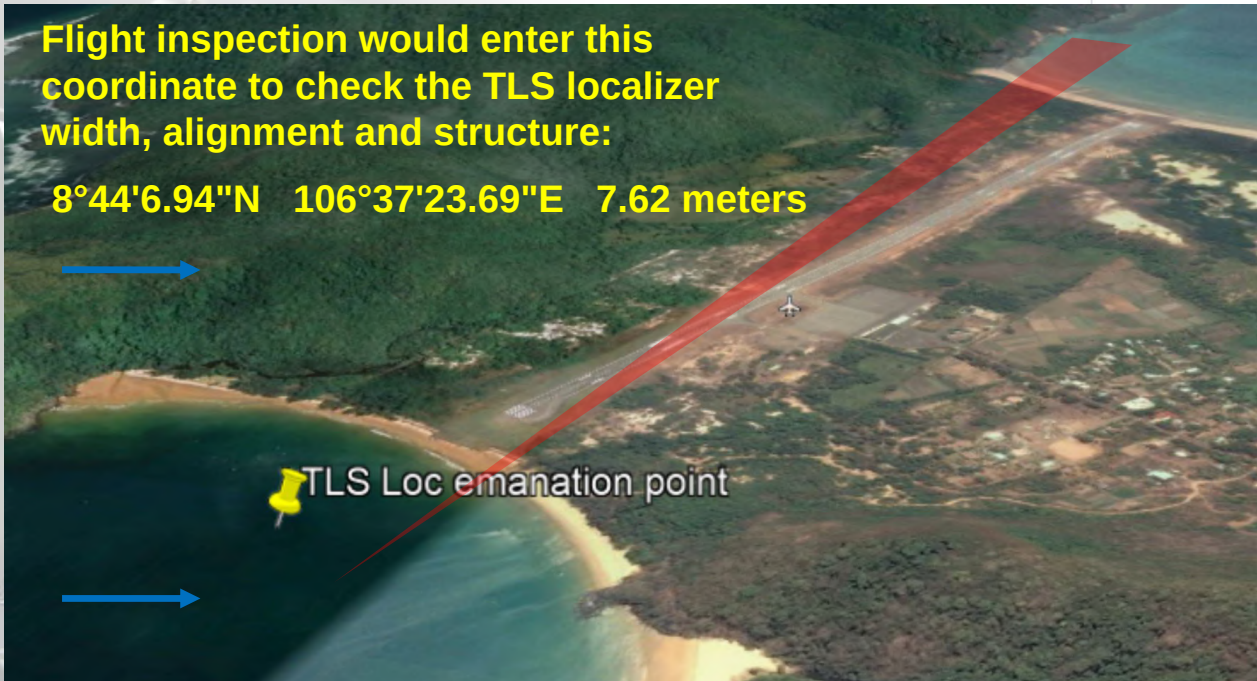
- The localizer transmitted by TLS appears to emanate from the traditional ILS localizer location



Flight inspection point for TLS localizer

Flight inspection would enter this coordinate to check the TLS localizer width, alignment and structure:

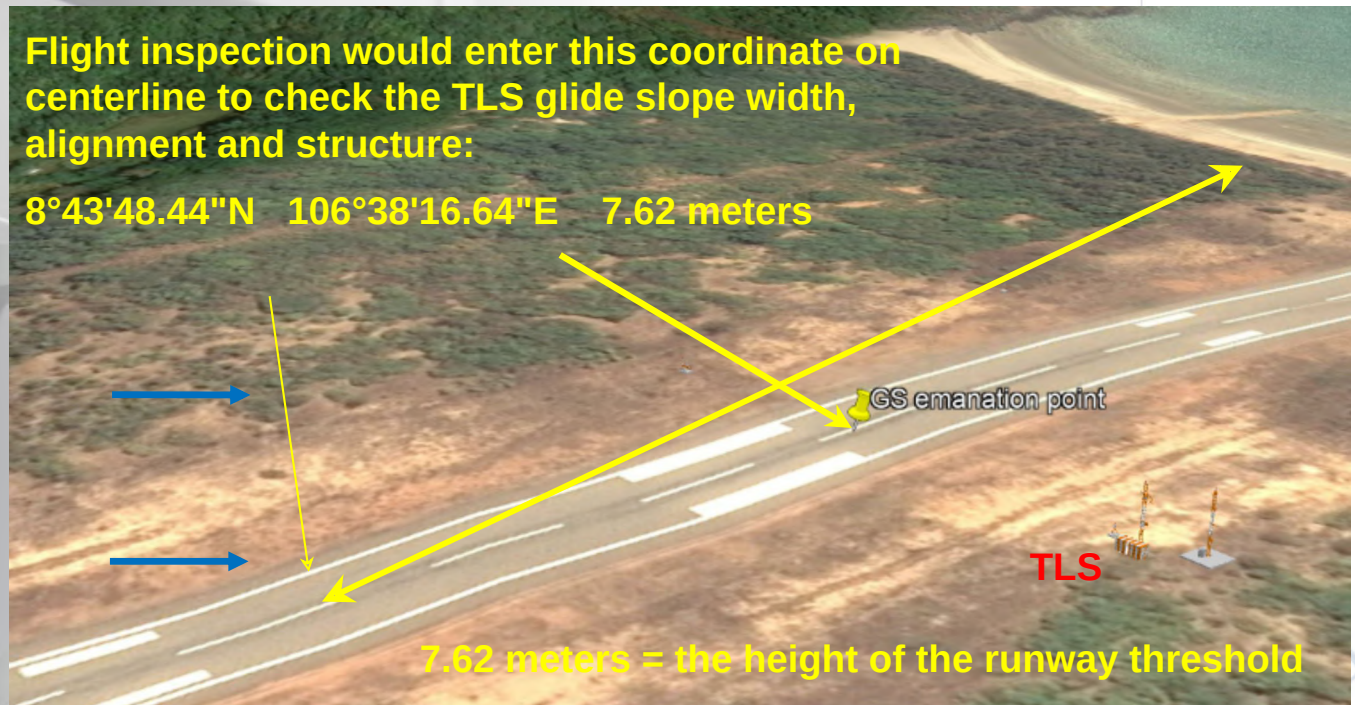
8°44'6.94"N 106°37'23.69"E 7.62 meters



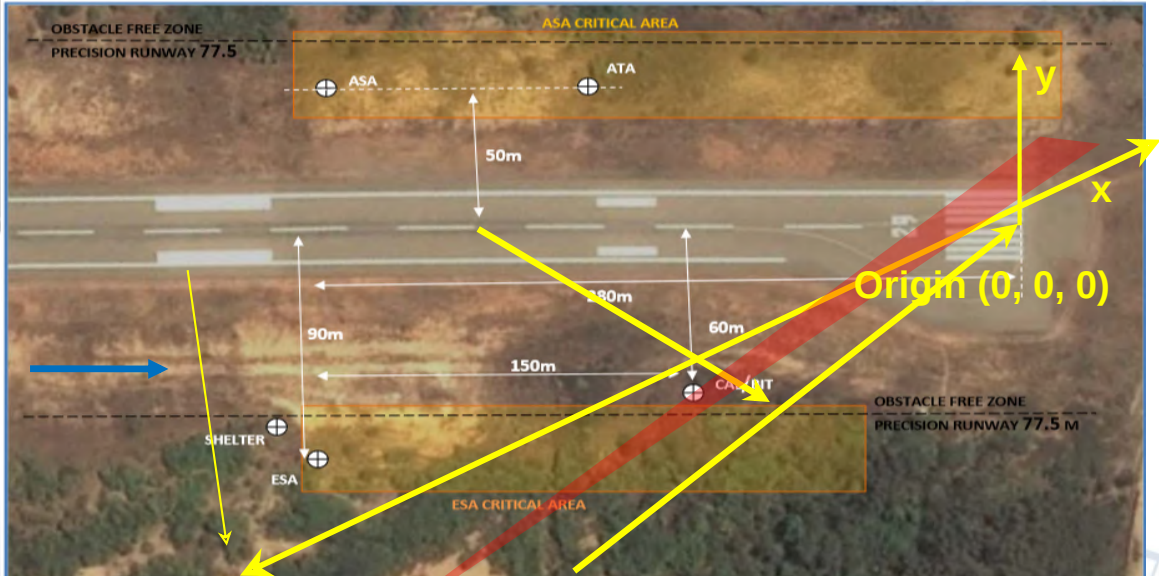
Flight inspection point for TLS localizer

Flight inspection would enter this coordinate on centerline to check the TLS glide slope width, alignment and structure:

8°43'48.44"N 106°38'16.64"E 7.62 meters



Survey of TLS location in relationship to RWY Threshold



The runway threshold is used as the TLS coordinate system zero reference point and the runway centerline is the reference orientation. All of the TLS antennas are surveyed relative to threshold and entered into the TLS maintenance interface as part of the installation process.

Configuration of the virtual localizer emanation point



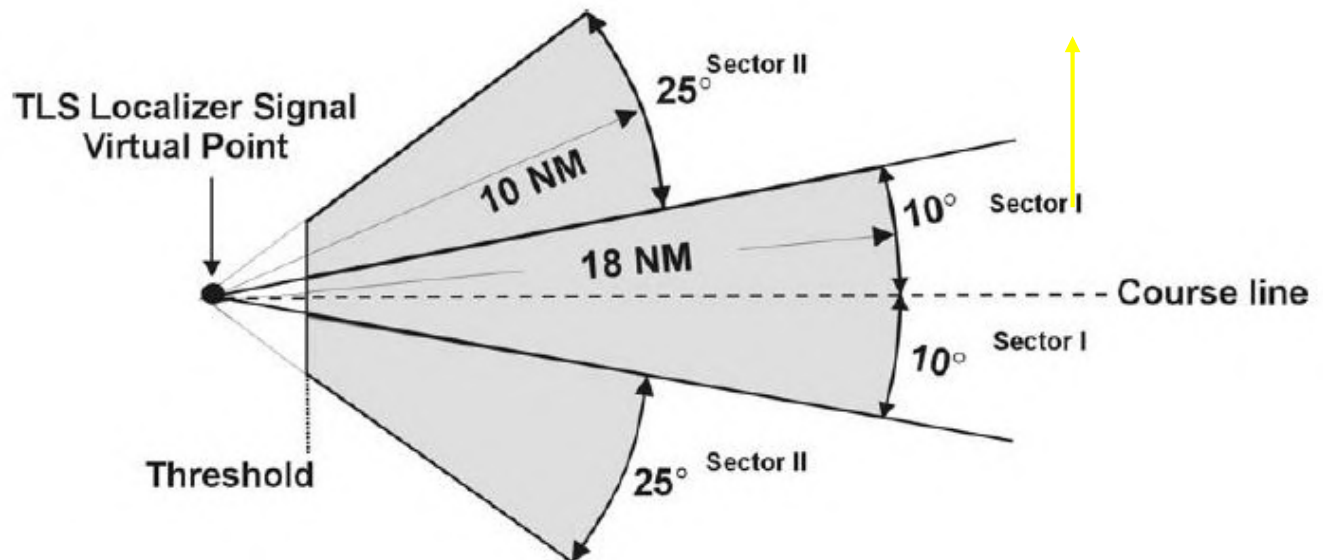
To achieve a 6 degree localizer course width, with a 210 meter width at threshold, the TLS localizer virtual emanation point is configured to be at the x coordinate of -2003.5 meters and y coordinate of 0.0 and z coordinate of 0.0.

Configuration of the virtual glide slope emanation point

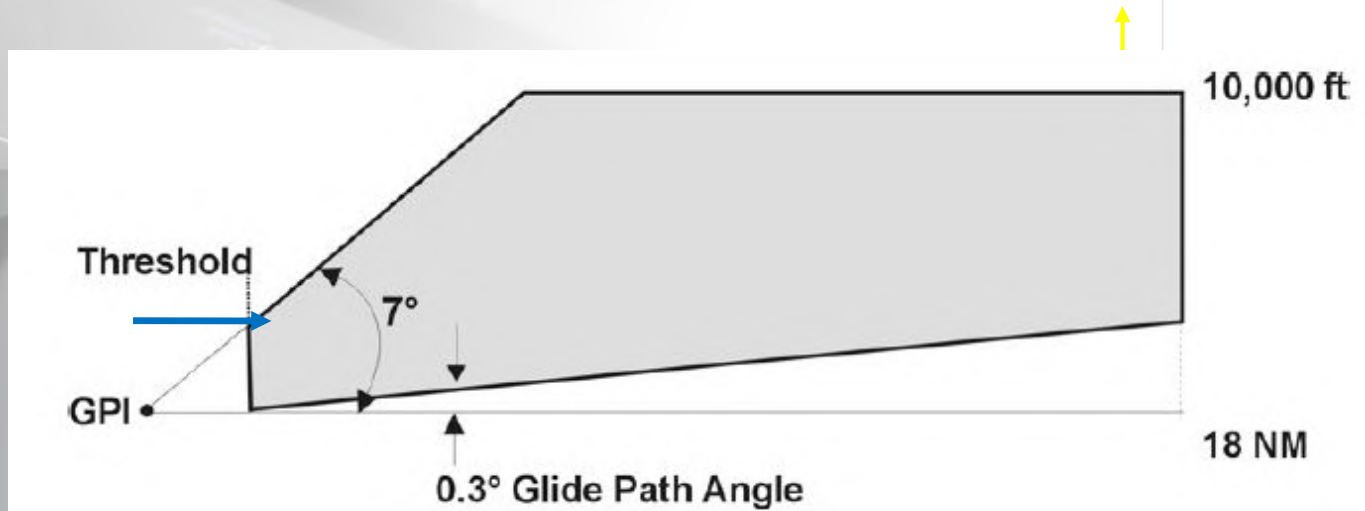


To achieve a 15 meter threshold crossing height and 3 degree glide slope, the TLS glide slope virtual emanation point is configured to be at the x coordinate of -286.2 meters and y coordinate of 0.0 and z coordinate of 0.0.

Service volume



Service volume

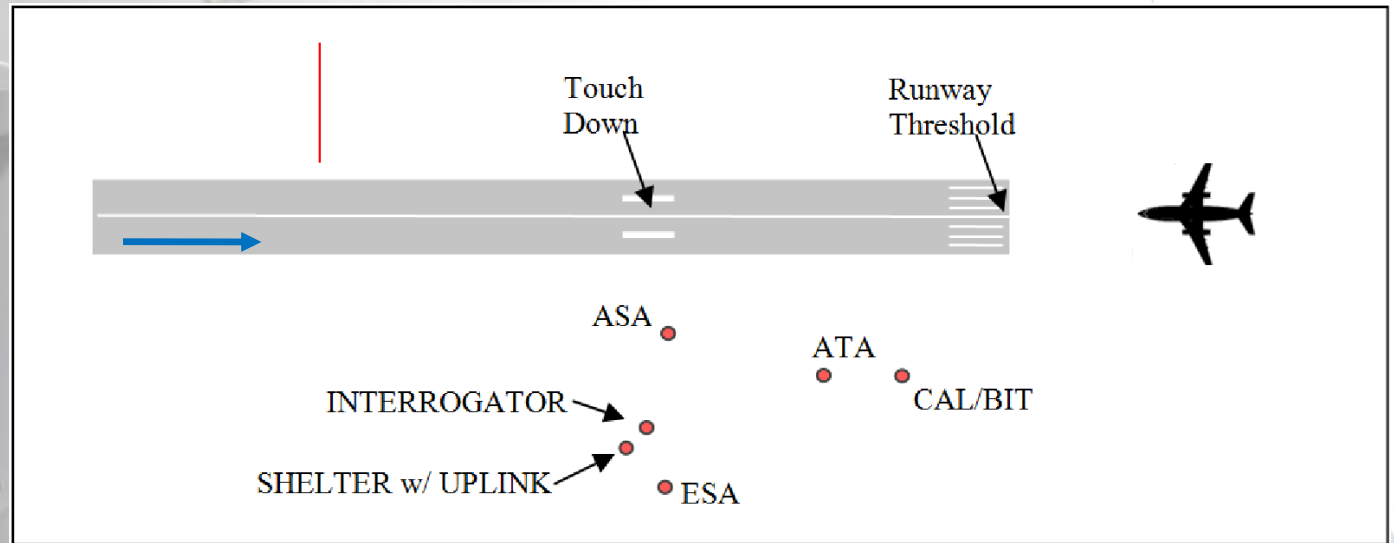


Technical contents

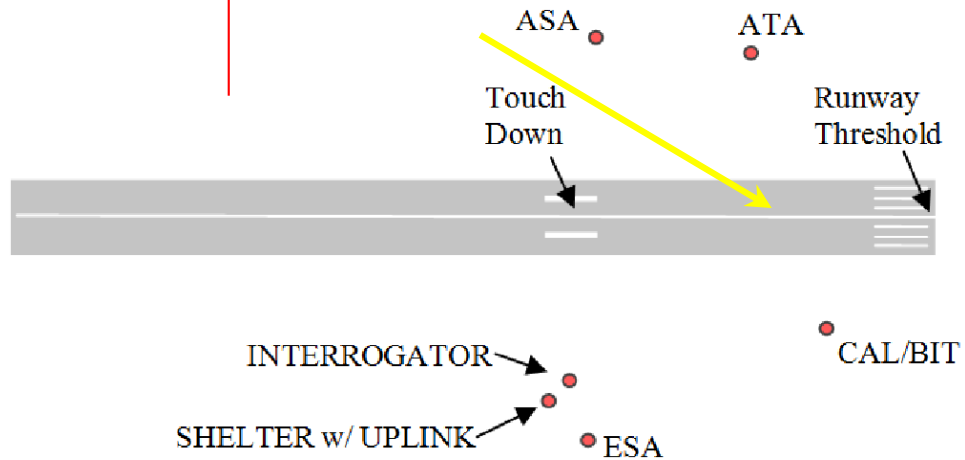
- ✓ Introduction to TLS
- ✓ Block diagram
- ✓ TLS components
- ✓ Virtual emanation points
 - Siting criteria
 - Transportable TLS
 - Theory (very briefly)
 - Survey procedure
 - Configuration of the integrity monitor
 - Calibration
 - Flight verification



Sited on one side of runway



Sited both sides of runway



Siting criteria

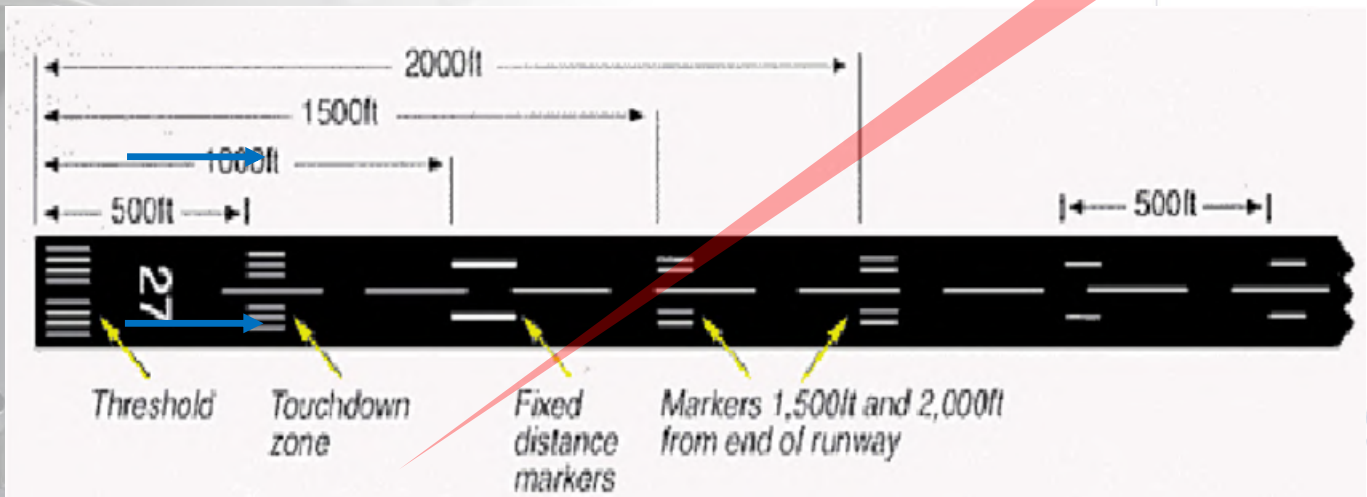
- See TI manual
- Summary table for TI manual shown below

Table 2-1 Summary of key siting parameters

	Minimum	Standard	Maximum
ASA w.r.t Touch Down (TD)	TD-100m	TD-50m	TD+75m
ASA to ESA separation	50 m (option 30 m)	100 m	No spec
ESA to Cal/BIT separation	150 m	150 m	No spec
ASA to ATA	50 m	100	150 m
ESA to ATA	50 m	100	150 m
ATA to Cal/BIT	20 m	60 m	No spec
Interrogator to shelter	11 m	12 m	13m
ASA to ESA angle from perpendicular to centerline	0 degrees	5 degrees	10 degrees
ASA to centerline	50m or as OFZ permits	60 m	70m
ESA to centerline	As permitted by local regs	90 m	200 m

TLS siting exercise

- Interactive: go through the exercise of thinking about where TLS components would be placed next to this runway
- (note: stock image shown is reversed from runway number and North up orientation)



Technical contents

- ✓ Introduction to TLS
- ✓ Block diagram
- ✓ TLS components
- ✓ Virtual emanation points
- ✓ Siting criteria
-  Transportable TLS
- Theory (very briefly)
- Survey procedure
- Configuration of the integrity monitor
- Calibration
- Flight verification

Transportable TLS Features

In addition to TLS features, TTLS provides:

- Transportability
- All components and power self contained
- Transportable by C-130
- Deployed in 10 hours by 3 Technicians
- Multiple displays
- controllers work independently guiding multiple aircraft

Transportable TLS for Joint Aviation Command

- Some features of the TTLS



Freightliner truck will be delivered under this contract

- Freightliner M2 106
 - 33,000 GVW,
 - 330HP Cummins and Allison Automatic,
 - Terra drive systems,
 - integrated container loading ramp,
 - painted flat tan



UAE System Deployed to Assab

- Load as a single unit or two pieces on smaller transport aircraft



Four remote control ATC work stations

Sunlight readable touch screen panels

Fully independent function so up to four controllers can operate

- ☐ **Includes wireless bridge with 5,000 foot range to have RCU remote from TTLS**



Quick connect power fiber cables



Dual Mitsubishi 30,700 Btu/Hr



- Two independent units
- Two indoor (one pictured)
- Two outdoor compressors



Maintenance free support hardware

- Stainless steel and anodized aluminum antenna infrastructure



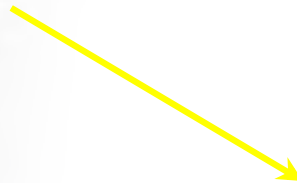
Kubota 11 KW diesel generator

- TLS includes UPS and batteries with 3 hours of backup
- Diesel generator automatic start when power is lost
 - Starts within 10 seconds of primary power loss
 - Automatically shuts off 5 minutes after power is restored



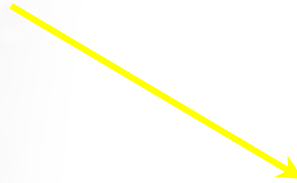
Technical contents

- ✓ Introduction to TLS
- ✓ Block diagram
- ✓ TLS components
- ✓ Virtual emanation points
- ✓ Siting criteria
- ✓ Transportable TLS
 - Theory (very briefly)
 - Survey procedure
 - Configuration of the integrity monitor
 - Calibration
 - Flight verification

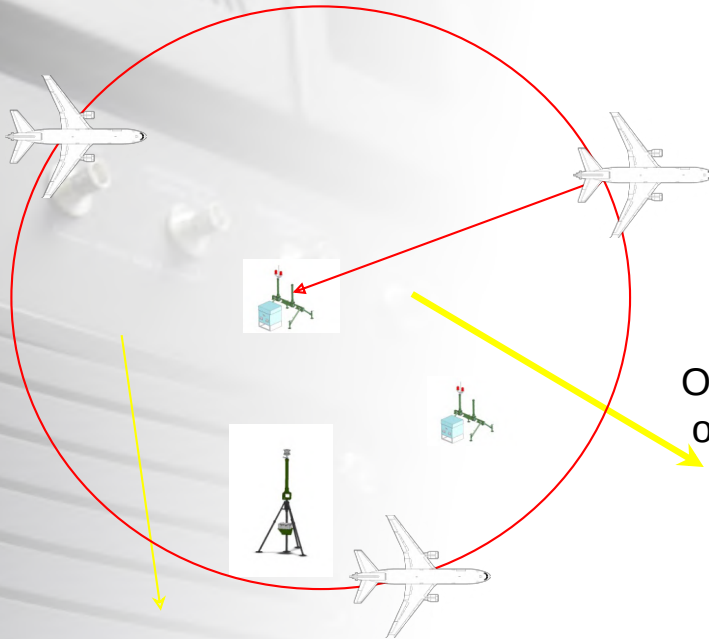


Theory

- TOA measurements

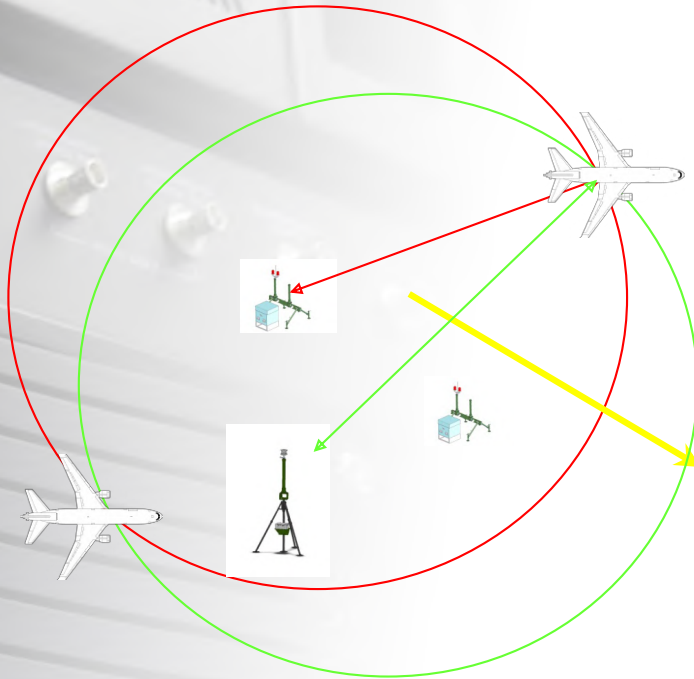


Time Of Arrival (TOA)



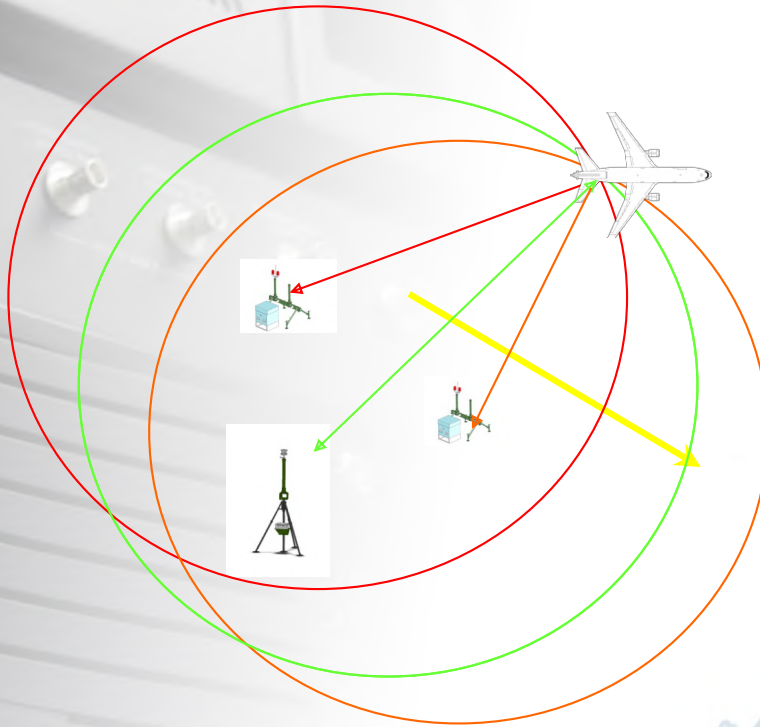
One TOA = range to the aircraft only

Time Of Arrival (TOA)



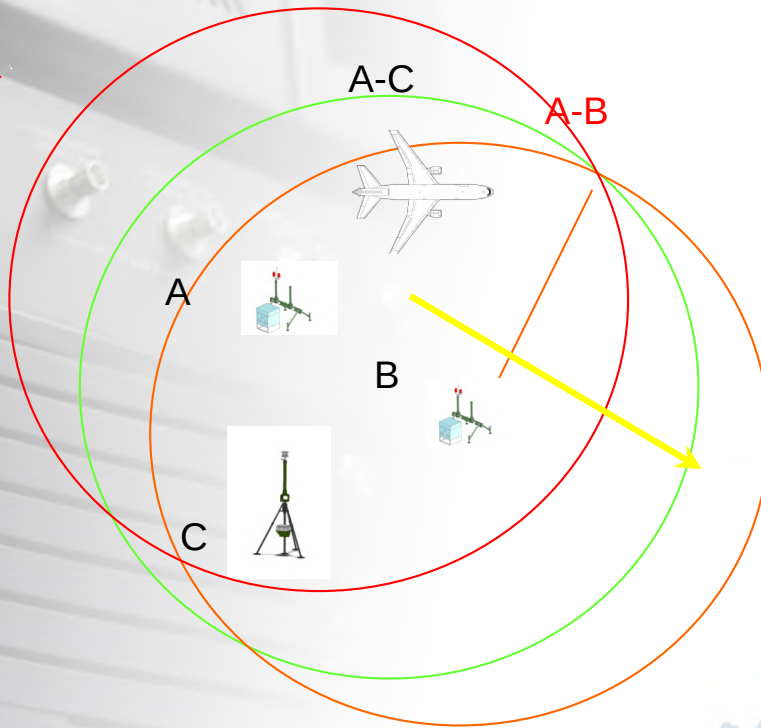
2 TOA = 2 positions

Time Of Arrival (TOA)

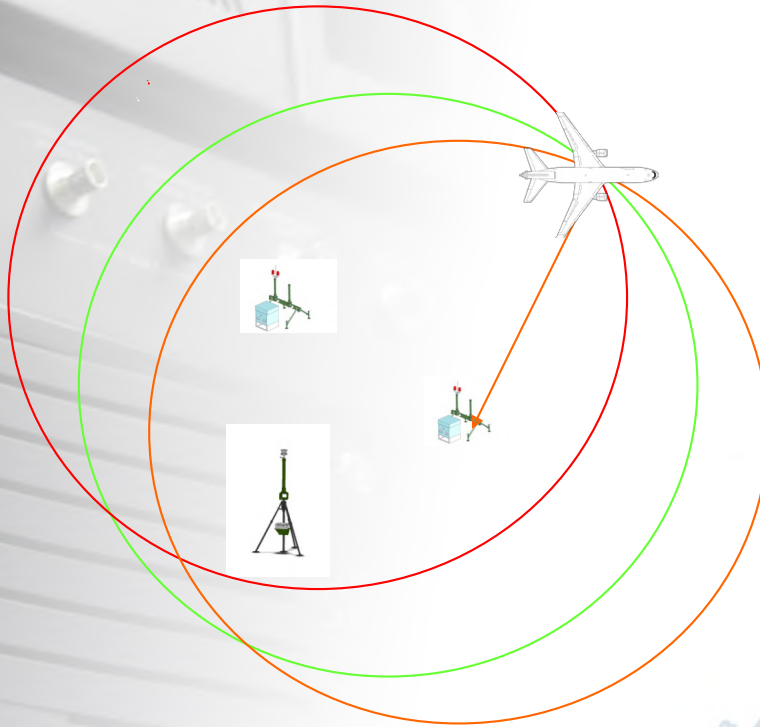


3 TOA = 1 position

Difference Time Of Arrival (DTOA)

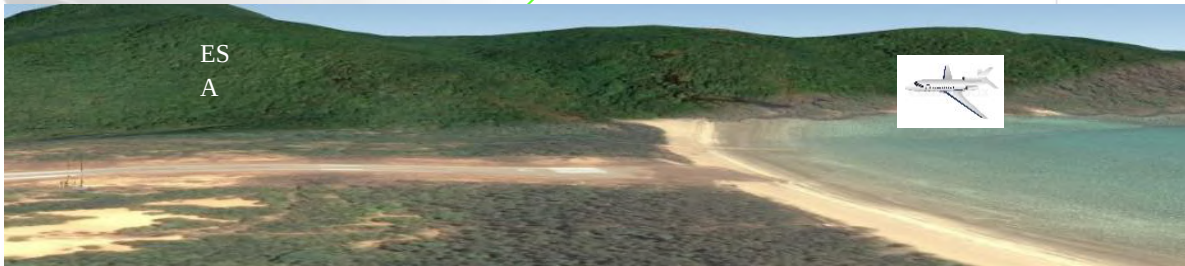


DTOA and TOA combine for better accuracy



Elevation Measurement Subsystem

- **Measures the aircraft height**



Elevation Measurement Subsystem



Two primary multipath mitigations:

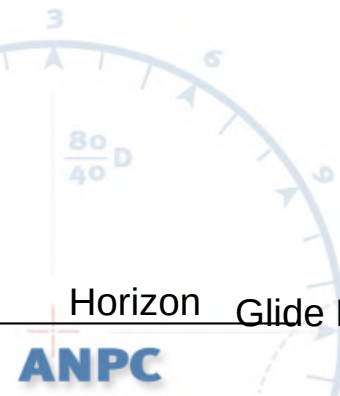
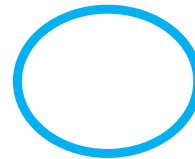
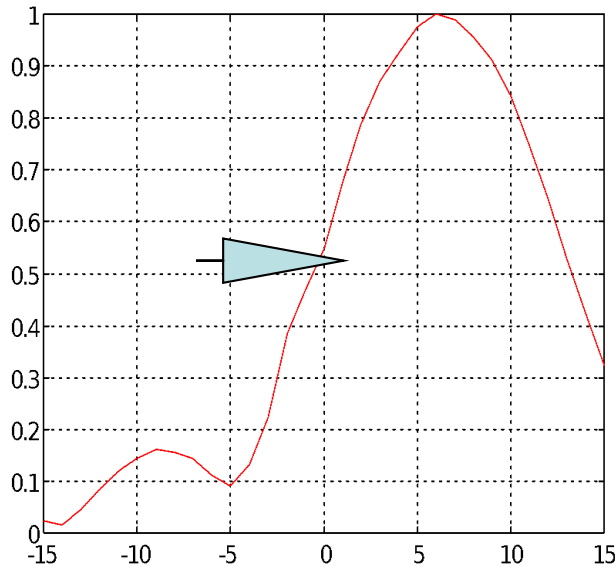
- Antenna vertical gain designed to reduce multipath amplitude
- Low frequency multipath compensated by data file



Antenna vertical gain selected to reduce the amplitude of ground-based multipath



**Gain of up to 12 dB
above the signal level
of the multipath**

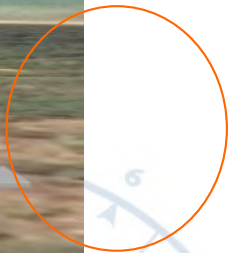
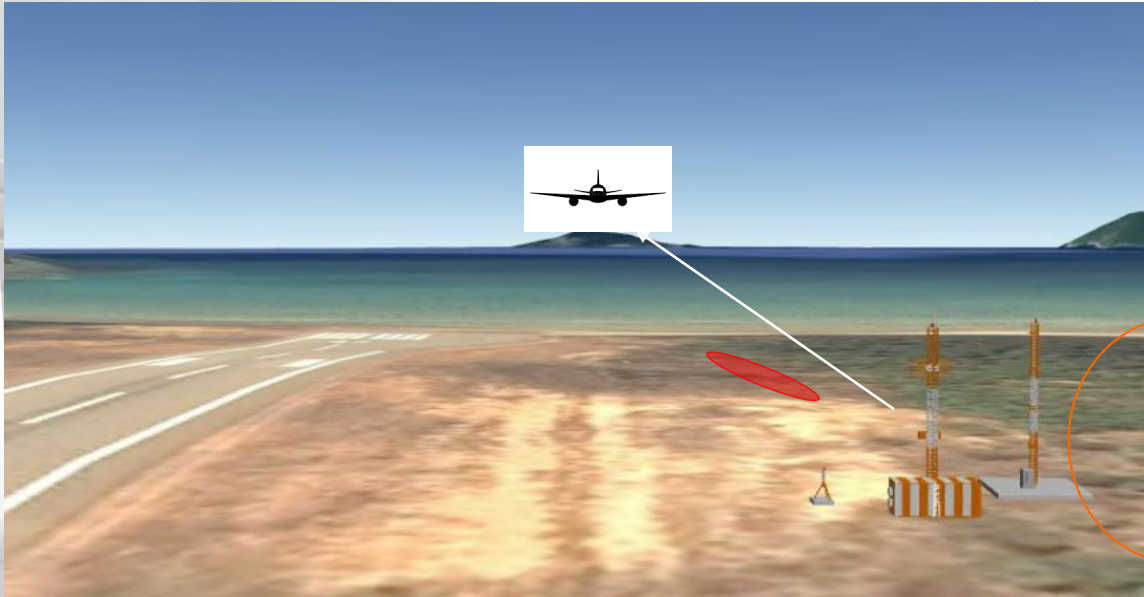




The uplink

- Base station shelter houses transmitters and CPU's
- VHF and UHF uplink are composite signals and therefore, the Difference Depth Modulation (DDM) is immune to multipath





Monitoring of critical parameters

- TLS has been designed in accordance with principles and airworthiness requirements of Aerospace Recommended Practice (ARP) 4754
- System failure modes have been eliminated, mitigated and reduced to an acceptable level by incorporating:
 - Built In Test (BIT)
 - Redundant systems
 - Level B design assurance

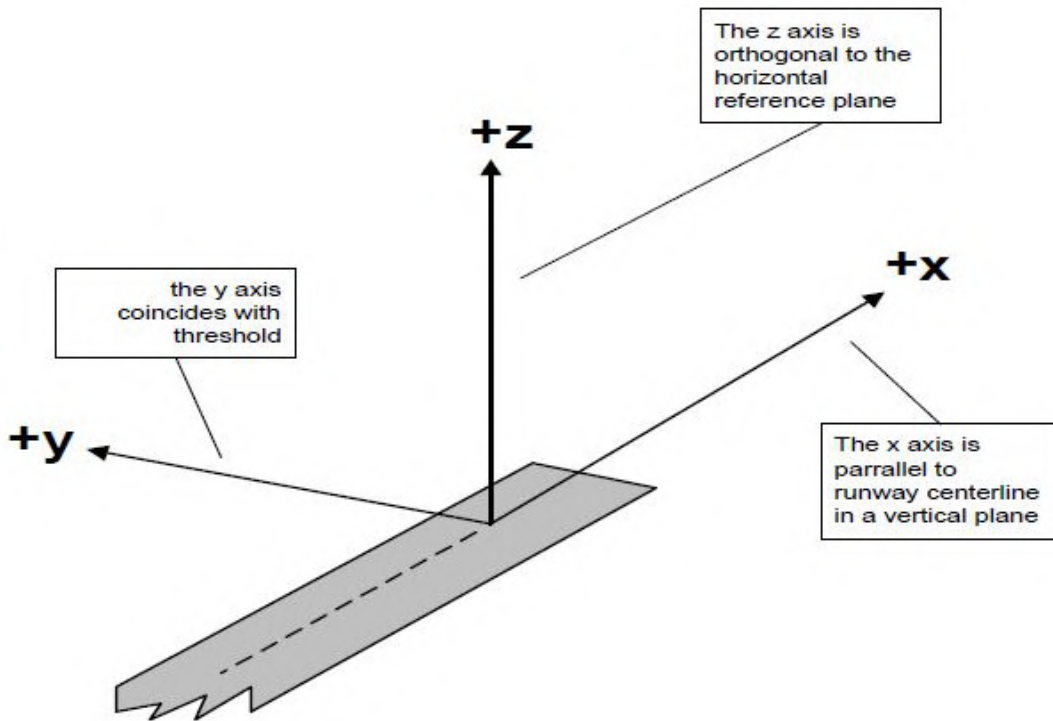


Technical contents

- ✓ Introduction to TLS
- ✓ Block diagram
- ✓ TLS components
- ✓ Virtual emanation points
- ✓ Siting criteria
- ✓ Transportable TLS
- ✓ Theory (very briefly)
- Survey procedure
- Configuration of the integrity monitor
- Calibration
- Flight verification



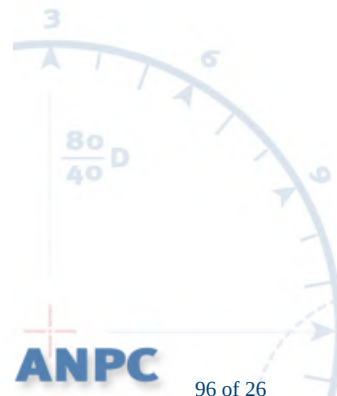
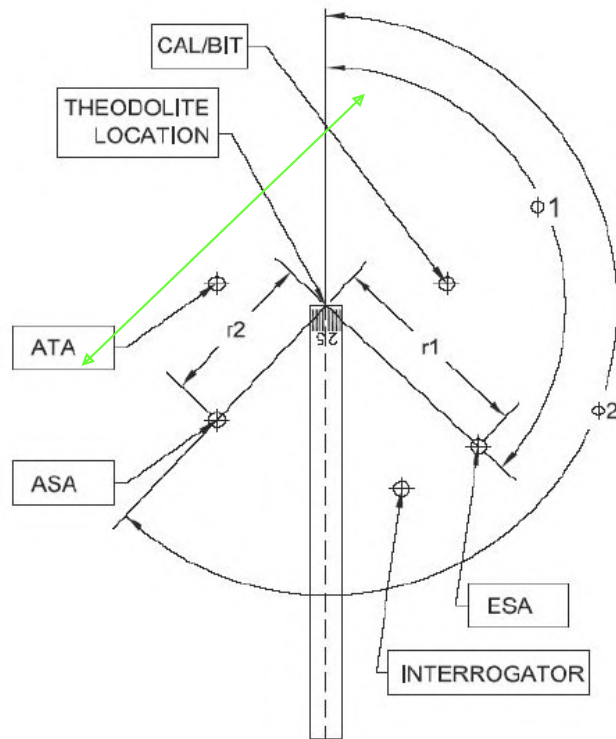
Survey procedure for TLS



Survey procedure for TLS

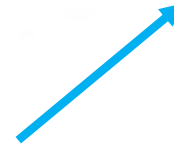


Survey procedure for TLS



Technical contents

- ✓ Introduction to TLS
- ✓ Block diagram
- ✓ TLS components
- ✓ Virtual emanation points
- ✓ Siting criteria
- ✓ Transportable TLS
- ✓ Theory (very briefly)
- ✓ Survey procedure
 - Configuration of the integrity monitor
 - Calibration
 - Flight verification



Configure GTU's



The Integrity monitor

- The Integrity Monitor has several components:
 - Built in Test (BIT) verifies that key performance parameters are within tolerance
 - BIT is active during System mode Track and during Track State
 - The tracking subsystem and uplink subsystem are monitored
- Key performance parameters include AOA phase, amplitude, frequency, DDM, SDM transmit signal level, P1, -p2 and p3 spacing, amplitude and pulse width
- Alarms and Alerts
 - Alert indicates a parameter close to being out of tolerance – system still operational
 - Alarm indicates a parameter out of tolerance – system non-operational
- Log files are generated for troubleshooting

Basic BIT analysis

- Open esa.bit and asa.bit with MultiBIT tool
- Look for correlations or complete lack of data
- Lack of data could be an indication of no Cal/BIT signal



Configure the Integrity monitor

- Tune the CalBIT amplitude and TOA Jitter
- Archive to clear out the old data
- Run BIT 5 times
- Use “compute nominals” on the ATA, ASA and ESA bit files to automatically set monitor limits
- Set the TOA offsets for the ATA, ASA and ESA

Do this now using laptop version of TLS SW tools



Technical contents

- ✓ Introduction to TLS
- ✓ Block diagram
- ✓ TLS components
- ✓ Virtual emanation points
- ✓ Siting criteria
- ✓ Transportable TLS
- ✓ Theory (very briefly)
- ✓ Survey procedure
- ✓ Configuration of the integrity monitor
 - Calibration
 - Flight verification

Calibration



The Purpose of calibration



Align TLS measurements to track the aircraft



Align the TLS localizer with runway centerline



Align the TLS glide slope at the required angle



Calibration ...



Is completed after the system is in standby



Is completed prior to flight inspection



Calibration Training Summary

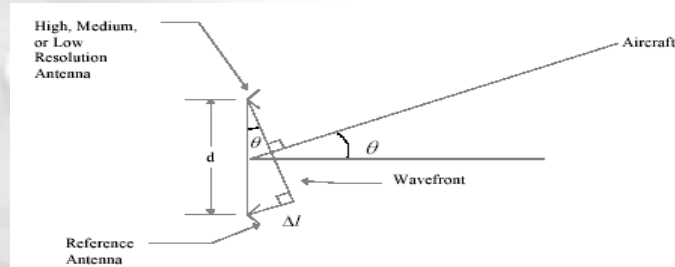
- During the next 30 minutes we will look more at
 - TLS measurement processing
 - Calibration data recording
 - Data processing
 - Using the Calibration tool
 - Understanding the measurement using the Look tool
 - Practical exercises with archive data that will match the experience in the field



Calibration Training Theory

□ TLS Measurement Processing

- AOA arrays measure differential phase of replies



- Δl measured in units of wavelengths

$$\lambda = \frac{c}{v} \approx \frac{299.8 \text{ m}/\mu\text{s}}{1090 \text{ MHz}} = 0.2750 \text{ m}$$

- Cycle size of an antenna pair

$$\theta = \sin^{-1} \left(\frac{\lambda}{d} \right)$$

- Measurement digitized into 256 counts (8 bits)

Calibration Training Theory

- Cycle sizes for TLS

- Azimuth

- Low: 21.0° $0.08^\circ / \text{count}$

- High: 5.2° $0.02^\circ / \text{count}$

- Elevation

- Low: 7.0° $0.027^\circ / \text{count}$

- Med: 3.5° $0.014^\circ / \text{count}$

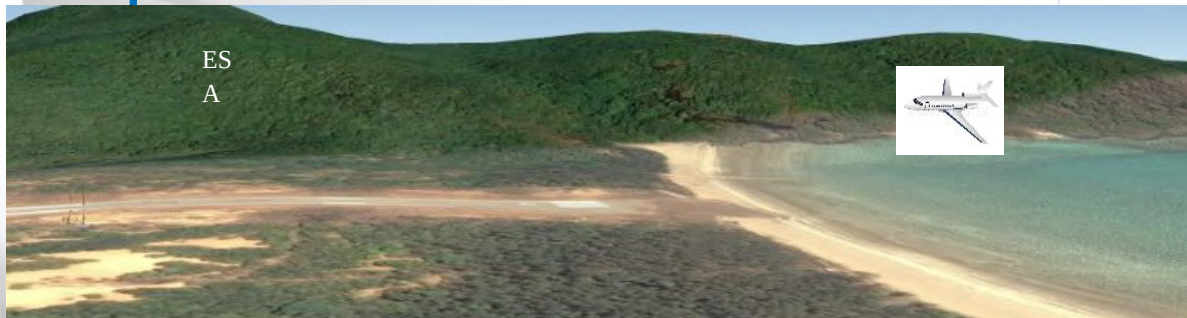
- High: 2.33° $0.009^\circ / \text{count}$

Calibration Training Theory

- Multipath

Short Path: Near field Fresnel zone bounce arrives out of sync

Long Path: Results in a trailing pulse, not relevant



Calibration Training Theory

- Three TLS subsystems are calibrated

TOA for the ATA, ASA and ESA

- Range and azimuth Track of each aircraft
- ASA Low resolution cycle selection
- Localizer alignment at threshold

ASA

- The Track derived from the ASA high res phase determines the localizer alignment
- Therefore the high res phase needs to be aligned with the runway

ESA

- The Track derived from the ASA high res phase determines the localizer alignment
- Therefore the high res phase needs to be aligned with the runway
- Elevation angle vs error curve removes “out of synch” error due to multipath

Calibration Training Data Collection

- Prerequisites
 - TLS properly configured
 - TLS in Standby (exclusive of RCU)
 - FTM is correctly located and operational
 - Best available cal.dat installed
 - Generate from monitor limits
 - Offset shift from old cal.dat
 - Test Profiles created and briefed
 - Schedule the aircraft
 - Brief the pilot



Calibration Training Data Collection

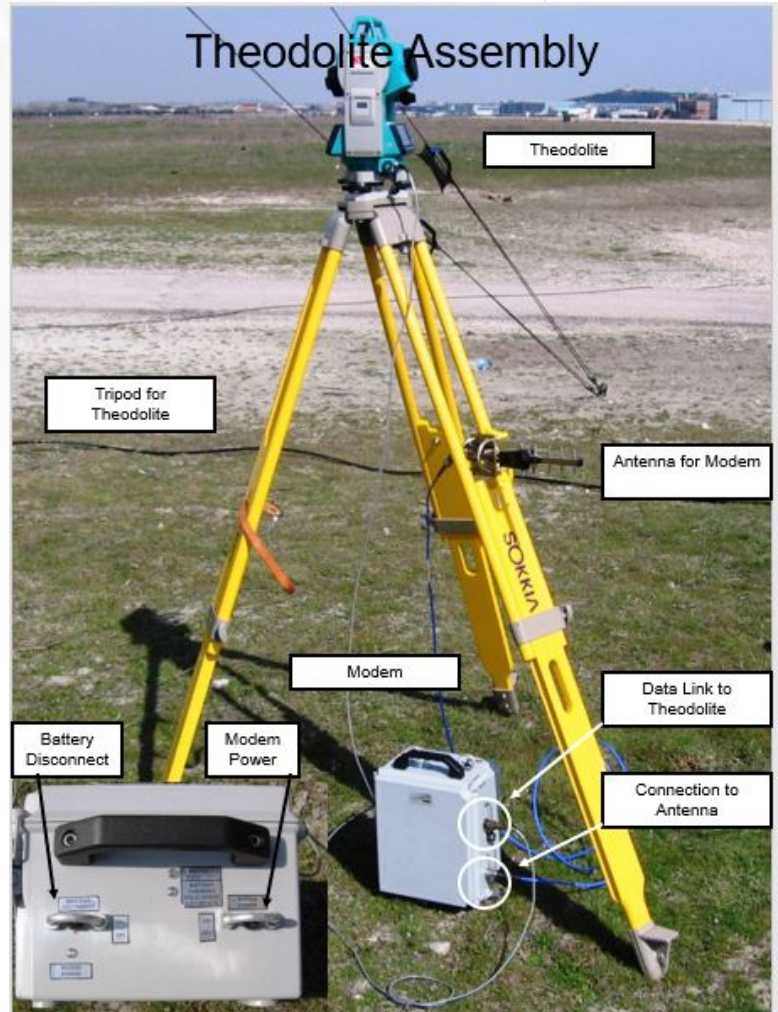
- Turn to the TI manual 020-00071 Calibration paragraphs on:
 - o Flight profiles,
 - o Testing the FTM link
 - o Briefing the pilot,
 - o Collecting the data

<#>



Theodolite part of FTM

- Important to position the theodolite so that the glide slope passes through the eyepiece



Replay of the flight : level crossing

- Replay from the training archive:
Glide slope level crossing



Calibration

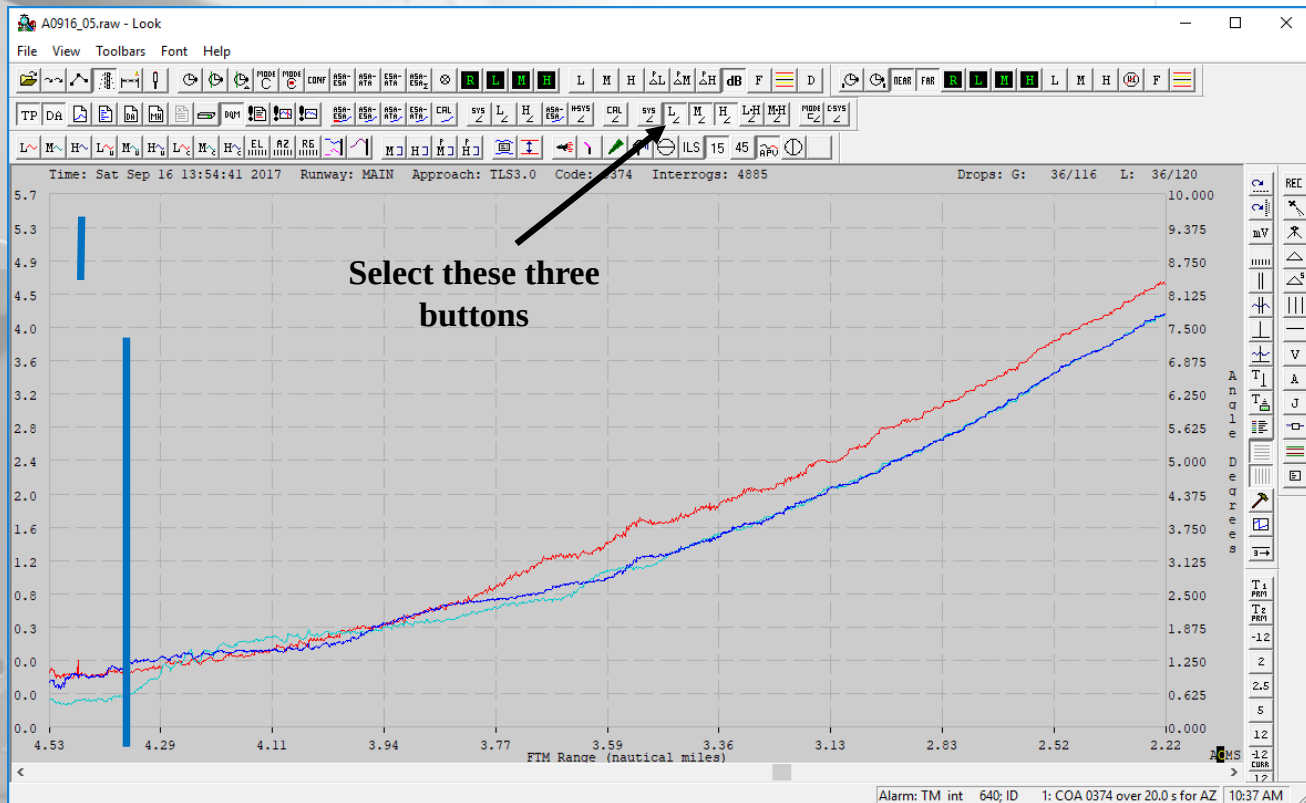


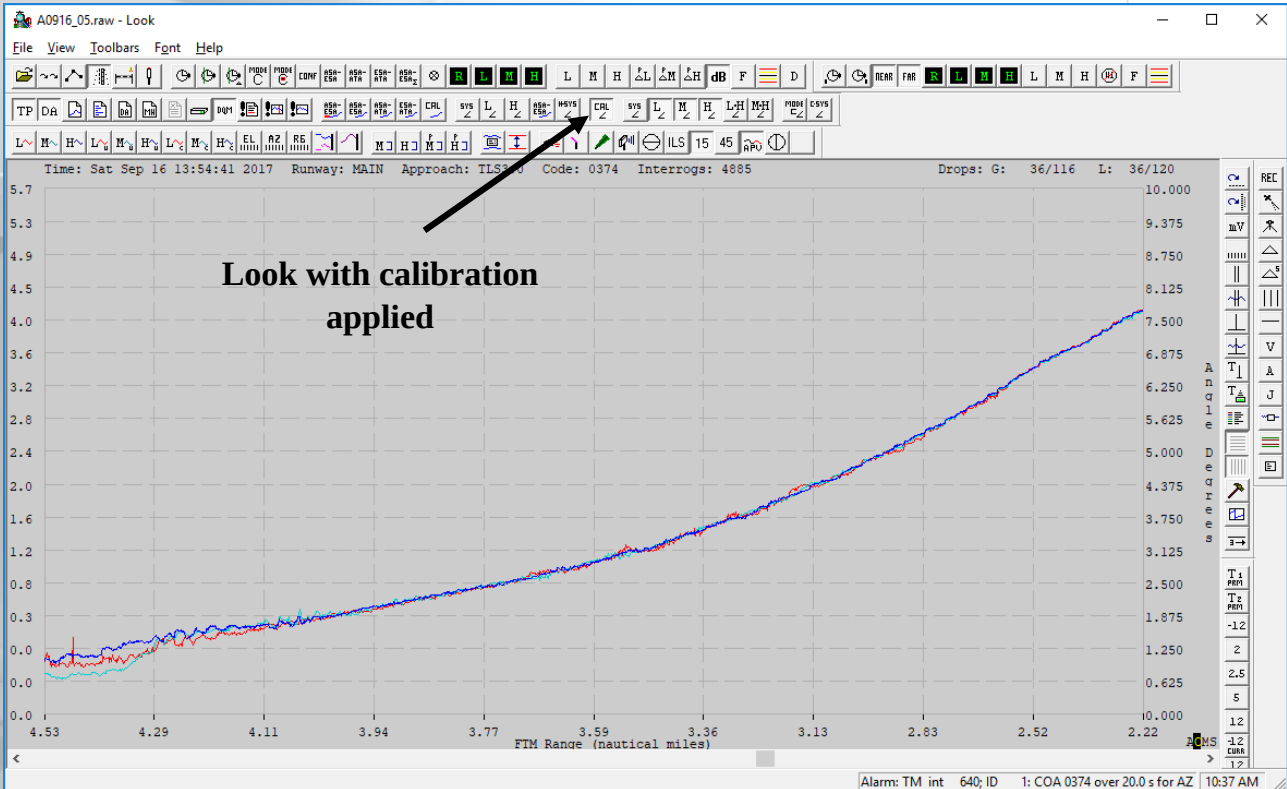
Calibration

- Now ... check your work
- Use the look tool to make sure the measurements are aligned
 - Double click on the raw file with the level crossing
 - Select the calibrated measurement buttons for low, medium and high res measurements
 - Verify that the measurements are with $\frac{1}{4}$ to $\frac{1}{2}$ degrees of each other
- If so, you're done!
- Else go back to the cal tool and make further adjustment to the line.
- Copy cal.dat to the base CPU's.



Calibration



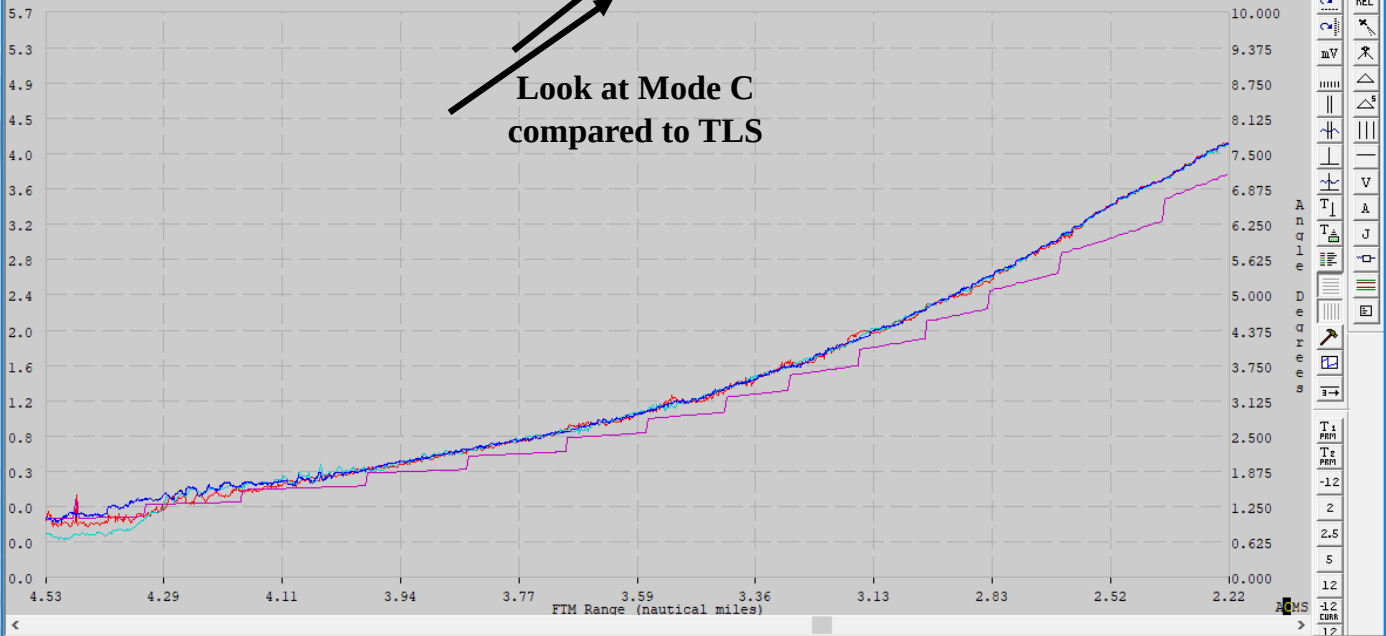


File View Toolbars Font Help



Time: Sat Sep 16 13:54:41 2017 Runway: MAIN Approach: TLS3.0 Code: 0374 Interrogos: 4885

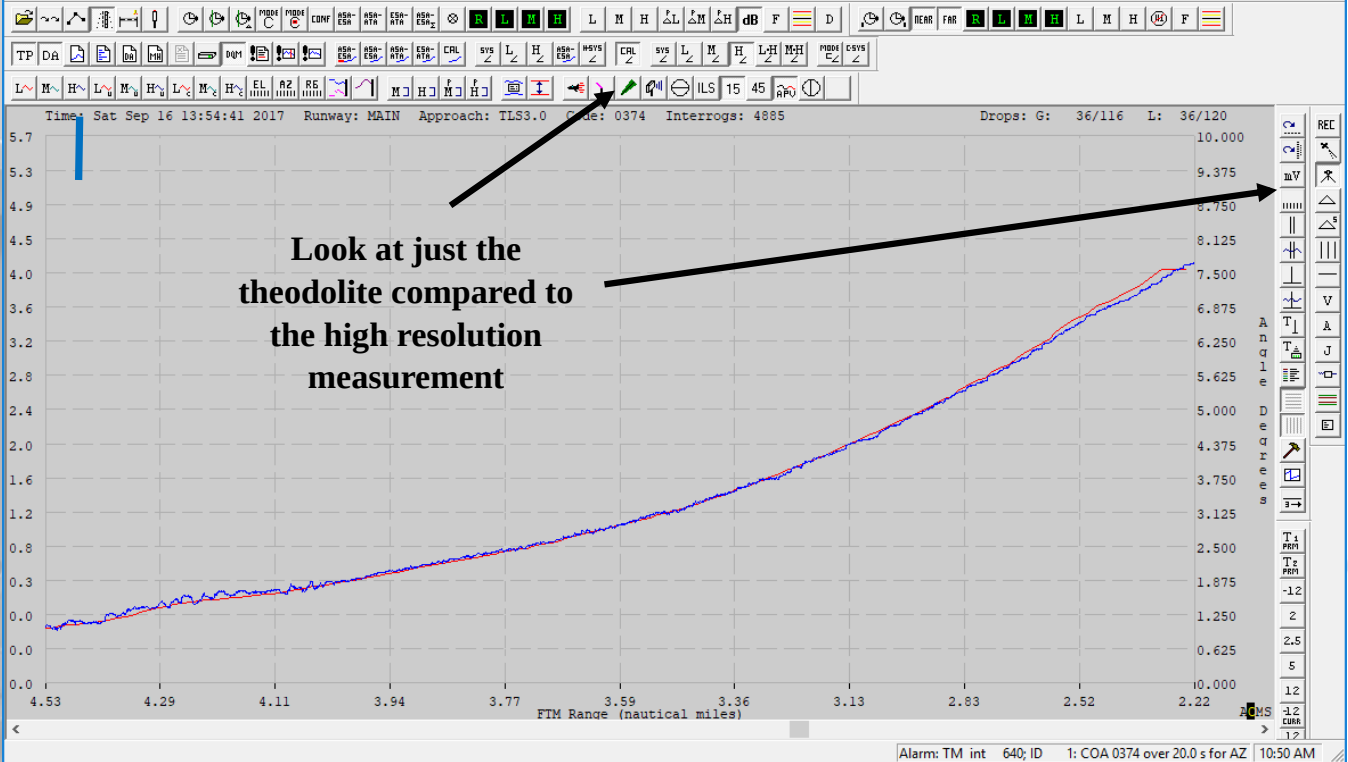
Drops: G: 36/116 L: 36/120



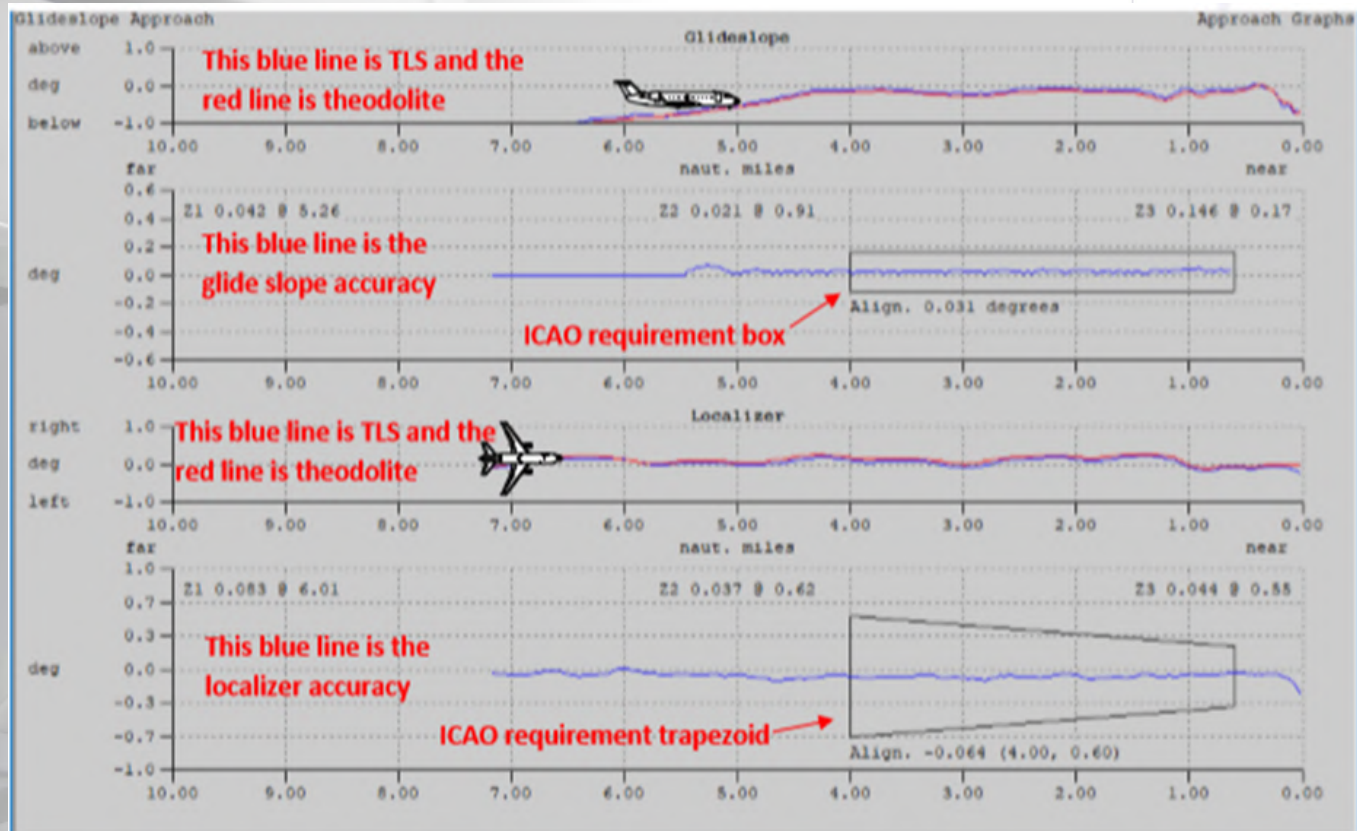
Alarm: TM int 640; ID 1: COA 0374 over 20.0 s for AZ 10:47 AM

 A0916_05.raw - Look

File View Toolbars Font Help



Glide Slope and localizer verification using FTM



Replay of the flight : level crossing

- localizer orbit to verify the service volume coverage
- No theodolite record is needed
- Replay from the training archive:
 - Localizer orbit 8nm
 - Localizer orbit at 22 nm



Calibration

- TLS is now ready for flight inspection
- Flight inspection will provide final adjustments to:
 - Glide Path width
 - Localizer width
 - Glide path alignment
 - Localizer alignment



Replay of a flight inspection

- Replay from the training archive:
 - Localizer orbit 5nm
 - Localizer orbit 22nm
 - Glide slope level crossing
 - Below path clearance
 - Glide slope left edge, glide slope right edge
 - On course and on path approach



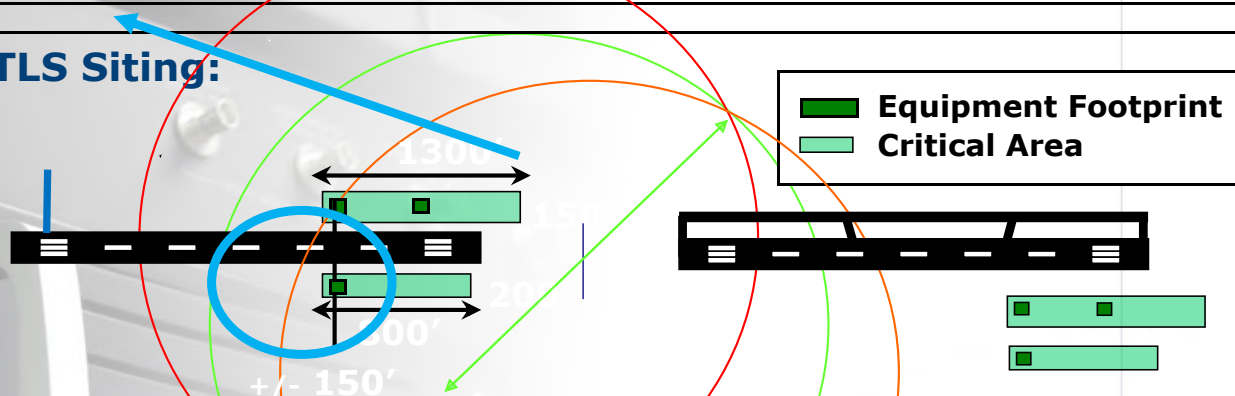
Technical contents

- ✓ Introduction to TLS
- ✓ Block diagram
- ✓ TLS components
- ✓ Virtual emanation points
- ✓ Siting criteria
- ✓ Transportable TLS
- ✓ Theory (very briefly)
- ✓ Survey procedure
- ✓ Configuration of the integrity monitor
- ✓ Calibration
 - Critical Areas
 - Maintenance
 - Operations
 - Flight verification



Critical areas

TLS Siting:

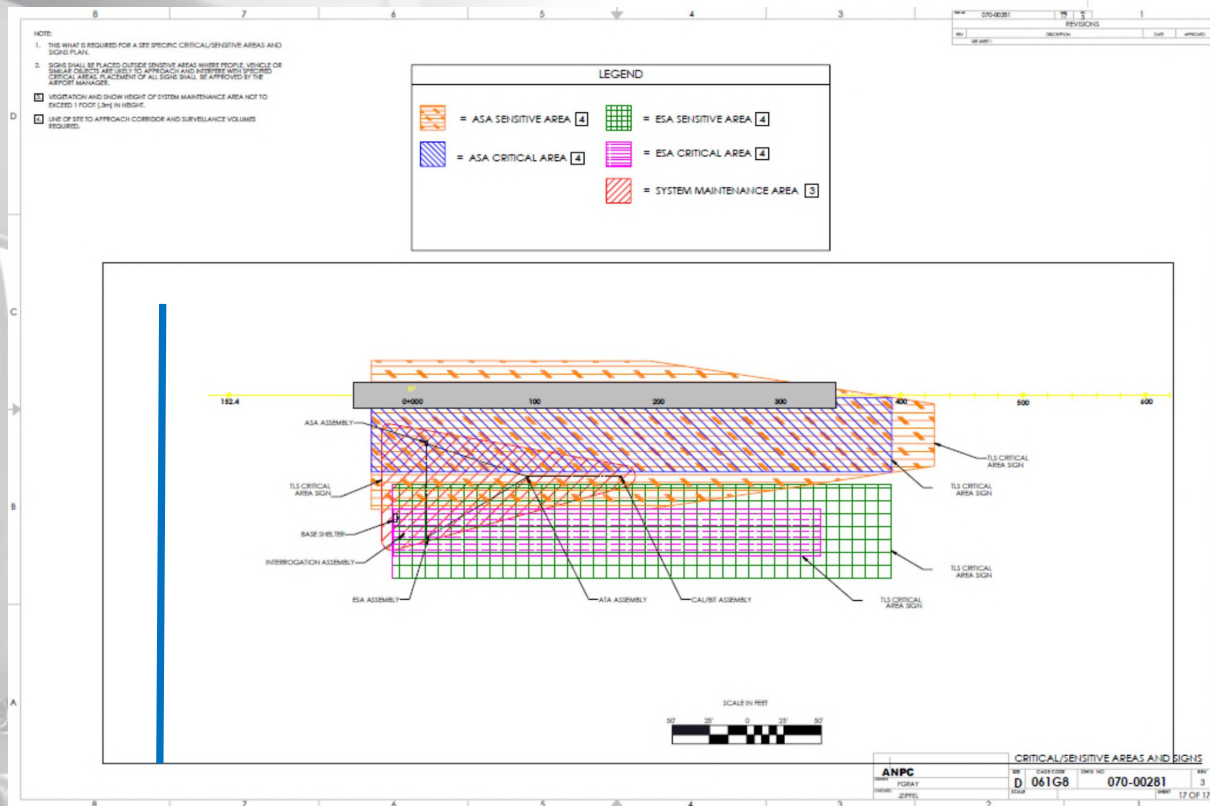


ILS frequently must be sited as shown:

Localizer on
centerline

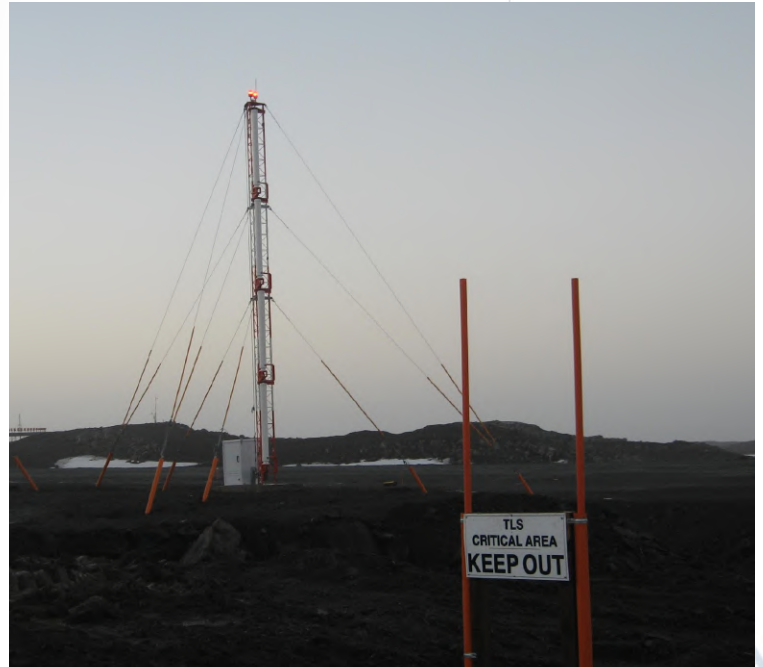


Critical areas



Critical area

- **Critical area sign – ATC is responsible for control of the area.**



Technical contents

- ✓ Introduction to TLS
- ✓ Block diagram
- ✓ TLS components
- ✓ Virtual emanation points
- ✓ Siting criteria
- ✓ Transportable TLS
- ✓ Theory (very briefly)
- ✓ Survey procedure
- ✓ Configuration of the integrity monitor
 -
- ✓ Calibration
- ✓ Critical Areas
 - Maintenance
 - Operations
 - Flight verification



Maintenance

020-00074 Rev. A

TLS Maintenance Manual

08/01/2012

Table 3-1 TLS Periodic Maintenance Schedule

Maintenance Item	Standards and Tolerances	Procedure #	Month											
			1	2	3	4	5	6	7	8	9	10	11	12
M-1 Archive of Data Files	Table 4-2, -3, -4, -5	5.3	X	X	X	X	X	X	X	X	X	X	X	X
M-2 RCU Functions Check and Monitor Shutdown Validation	Table 4-7	5.4												X
M-3 Maintenance of Critical Areas and Obstruction Lights	N/A	5.5	X	X	X	X	X	X	X	X	X	X	X	X
M-4 UPS Battery Check	Table 4-7	5.6						X						X
M-5 Air Filters	N/A	5.7			X			X			X			X
M-6 Check Guy Wire Tension	N/A	5.8						X						X
M-7 RSC	N/A	5-9												
M-8 Antenna System Maintenance	N/A	5.10												X
M-9 Clean and Inspect Equipment	N/A	5.11												X
M-10 Interrogation Transmitters Check	Table 4-2	5.12												X
M-11 Guidance Transmitter Check	Table 4-3 and 4-4	5.13												X
M-12 Cable repair and waterproofing	N/A	5.14	As Required											

Maintenance

- First – check the base log

020-00074 Rev. A

TLS Maintenance Manual

08/01/2012

Table 3-3 Sample Error Log

System error logs are saved in the Run Log folders of the base station computers.

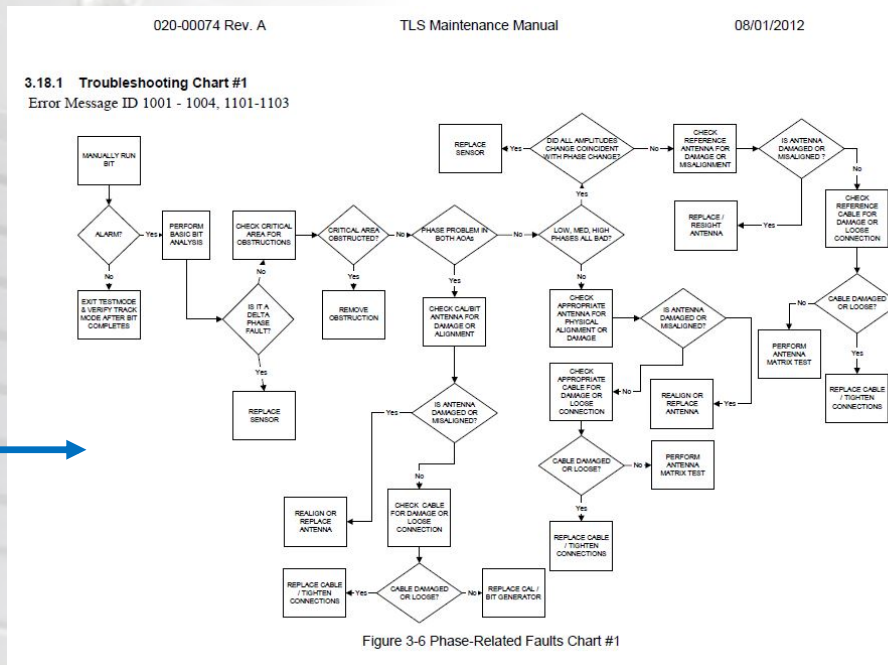
Date	Time	Type	Msg	LRU ID	SW ID	Message text
11/11/1997	11:10:15		INFO	0000	00	UL
11/11/1997	11:10:15		INFO	0000	00	UL
11/11/1997	11:10:15		INFO	0000	00	UL
11/11/1997	11:10:15		ALERT	3051	07	UL
11/11/1997	11:10:15		ALERT	3020	07	UL
11/11/1997	11:10:16		ALERT	3011	07	UL
11/11/1997	11:10:16		ALARM	3012	07	UL
11/11/1997	11:10:16		ALERT	3011	07	UL
11/11/1997	11:10:16		ALARM	3012	07	UL
11/11/1997	11:10:16		ALERT	3020	07	UL
11/11/1997	11:10:27		ALERT	3040	07	UL
11/11/1997	11:10:28		ALERT	3043	07	UL
11/11/1997	11:10:30		ALERT	3044	07	UL
11/11/1997	11:10:31		ALERT	3041	07	UL
11/11/1997	11:10:37		ALERT	3042	07	UL

START OF NEW LOG MESSAGES

UL--bit tone control bit mismatch
 UL--bit tone 0.188184 mv power out of range
 UL--bit localizer sdm 0.800000 out of range
 UL--bit localizer carrier power 0.188184 mv out of range
 UL--bit glideslope sdm 1.600000 out of range
 UL--bit glideslope carrier power 0.188184 mv out of range
 UL--bit GTU 5V 0.188184 out of range
 UL--bit GTU 5V 0.188184 out of range
 UL--bit GTU Localizer 24V 0.188184 out of range
 UL--bit GTU Glideslope 24V 0.188184 out of range
 UL--bit GTU 15V 0.188184 out of range
 UL--bit GTU -15V 0.188184 out of range

Maintenance

- Use the troubleshooting flow charts



Maintenance

- **Spares typically include**
 - AOA sensor electronics**
 - Cal/BIT**
 - The entire second rack of hot spare electronics**
 - Obstruction light**
- **We will talk in the class about the procedure for replacing each spare**
- **We will practice replacing each spare in the practical field portion of the training**

Technical contents

- ✓ Introduction to TLS
- ✓ Block diagram
- ✓ TLS components
- ✓ Virtual emanation points
- ✓ Siting criteria
- ✓ Transportable TLS
- ✓ Theory (very briefly)
- ✓ Survey procedure
- ✓ Configuration of the integrity monitor
- ✓ Calibration
- ✓ Critical Areas
- ✓ Maintenance
 - Operations
 - Flight verification



Remote Control Unit (RCU)

- Run DTLS



ADVANCED NAVIGATION & POSITIONING CORPORATION



Thanks for listening

489 N. 8TH ST. #203, HOOD RIVER, OREGON 97031
TELEPHONE 541.386.1746 WWW.ANPC.COM